TRANSLATIONAL SCIENCE

Characterised intron retention profiles in muscle tissue of idiopathic inflammatory myopathy subtypes

Handling editor Josef S Smolen

► Additional supplemental material is published online only. To view, please visit the journal online (https://doi.org/10.1136/ard-2023-225035).

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Received 21 September 2023
Accepted 23 January 2024
Published Online First
1 February 2024

ABSTRACT

Objectives Idiopathic inflammatory myopathies (IIMs) are a group of heterogeneous autoimmune diseases. Intron retention (IR) serves as an important post-transcriptional and translational regulatory mechanism. This study aims to identify changes in IR profiles in IIM subtypes, investigating their influence on proteins and their correlations with clinical features.

Methods RNA sequencing and liquid chromatography-tandem mass spectrometry were performed on muscle tissues obtained from 174 patients with IIM and 19 controls, following QC procedures. GTFtools and iREAD software were used for IR identification. An analysis of differentially expressed IRs (DEIs), exons and proteins was carried out using edgeR or DEP. Functional analysis was performed with clusterProfiler, and SPIRON was used to assess splicing factors.

Results A total of 6783 IRs located in 3111 unique genes were identified in all IIM subtypes compared with controls. IIM subtype-specific DEIs were associated with the pathogenesis of respective IIM subtypes. Splicing factors YBX1 and HSPA2 exhibited the most changes in dermatomyositis and immune-mediated necrotising myopathy. Increased IR was associated with reduced protein expression. Some of the IIM-specific DEIs were correlated with clinical parameters (skin rash, MMT-8 scores and muscle enzymes) and muscle histopathological features (myofiber necrosis, regeneration and inflammation). IRs in IFIH1 and TRIM21 were strongly correlated with anti-MDA5+ antibody, while IRs in SRP14 were associated with anti-SRP+ antibody.

Conclusion This study revealed distinct IRs and specific splicing factors associated with IIM subtypes, which might be contributing to the pathogenesis of IIM. We also emphasised the potential impact of IR on protein expression in IIM muscles.

WHAT IS ALREADY KNOWN ON THIS TOPIC

⇒ Intron retention (IR) is an important mechanism for post-transcriptional and translational regulation, influencing protein expression in cells.
⇒ Recently, IR has been implicated in the pathogenesis of autoimmune diseases, muscular dystrophies and myopathies but not in idiopathic inflammatory myopathy (IIM).

WHAT THIS STUDY ADDS

⇒ IIM subtype-specific differentially expressed IRs (DEIs) were associated with the pathogenesis of respective IIM subtypes.
⇒ Splicing factors YBX1 and HSPA2 exhibited the most changes in dermatomyositis and immune-mediated necrotising myopathy. The increased IR was associated with a reduced level of protein expression in IIM muscles.
⇒ IIM subtype-specific DEIs were correlated with clinical parameters and muscle histopathological features, including myofiber necrosis, regeneration and inflammatory cell infiltration. Additionally, certain IRs were associated with the levels of autoantibodies in IIM.

HOW THIS STUDY MIGHT AFFECT RESEARCH, PRACTICE OR POLICY

⇒ The IIM subtype-specific IRs and splicing factors may provide valuable insights into the understanding of pathogenesis, serve as new biomarkers, enable early classification diagnosis and represent promising therapeutic targets in the future for patients with IIM.

INTRODUCTION

Idiopathic inflammatory myopathies (IIMs) are a group of heterogeneous autoimmune diseases characterised by muscle weakness with non-suppurative inflammation, myositis-specific antibodies (MSAs) in serum and extramuscular manifestations such as skin rash and interstitial lung disease (ILD).[1] The prevalence of IIM estimates ranges from 2.9 to 34 patients per 100 000 persons all over the world.[2] Dermatomyositis (DM), immune-mediated necrotising myopathy (IMNM) and antisynthetase syndrome (ASS) are the more common subtypes of IIM.[1 2]

Although IIM shares many common characteristics, the presentation of the disease varies among the main subtypes and within MSA-defined subtypes.[3] The prominent features of DM include skin lesions like heliotrope rash and Gottron's papules; the presence of autoantibodies in serum such as anti-melanoma differentiation-associated gene 5 (MDA5), anti-transcription intermediary factor 1-gamma (TIF1γ), anti-Mi2, anti-nuclear matrix protein 2 (NXP2) or anti-small ubiquitin-like modifier-activating enzyme (SAE) antibodies; as well as the activation of type 1 interferon (IFN) pathway marked by the appearance of myxovirus resistance protein 1 (MX1) in DM muscles.[1 3 4] However, the manifestation of DM can differ among

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To cite: Xiao Y, Xie S, Li H-D, *et al*. *Ann Rheum Dis* 2024;**83**:901–914.

MSA-defined subtypes.[3] Notably, patients with anti-MDA5 positivity represent a unique phenotype of DM, characterised by typical cutaneous manifestations and an increased frequency of rapidly progressive ILD but minimal or no muscle damage.[1] Anti-TIF1γ+ DM typically carries a high malignancy risk. Anti-NXP2+ DM often presents with subcutaneous calcinosis and muscle ischaemia.[5 6] Anti-SAE+ DM is characterised by skin lesions with mild myopathic symptoms, while anti-Mi2+ DM frequently exhibits typical DM skin lesions and perifascicular necrosis in muscle biopsy.[7 8] In IMNM, characteristic features include severe muscle weakness, elevated levels of creatine kinase (CK) and the presence of anti-3-hydroxy-3-methylglutaryl-CoA reductase (HMGCR) or anti-signal recognition particle (SRP) antibodies in the serum, accompanied by scattered myofiber necrosis and macrophage infiltrates.[9] ASS may manifest various clinical symptoms, including myositis, ILD, mechanic hands, arthritis, Raynaud phenomenon, fever and the presence of antibodies against aminoacyl transfer RNA, such as anti-Jo-1, anti-EJ, anti-PL-7 and anti-PL-12.[10] Additional muscle pathological characteristics, such as perifascicular atrophy and perifascicular necrosis, are more commonly observed in DM but may also be present in ASS.[1] To date, effective treatment for IIM remains challenging. There is a critical need to understand the mechanisms of muscle damage in IIM and explore additional therapeutic targets.[1]

Intron retention (IR) is a well-conserved form of alternative splicing and a ubiquitous phenomenon in which an intron remains in the mature messenger RNA (mRNA) in cells. Even though nearly 95% of human genes undergo alternative splicing including IR, for a long time, IR had been systematically underestimated.[11–13] In fact, only 30%–40% of the variance in protein abundance could be explained by mRNA abundance in cell lines because of post-transcriptional, translational regulation and protein degradation.[13 14] IR can regulate gene expression via nonsense-mediated decay (NMD), mRNA detention in the nucleus or cytoplasm or alteration of translation efficiency or transcript stability, resulting in biological dysfunctions.[11 12] Additionally, the functions of IR can be influenced by factors such as their abundance, length, location and structure.[15–17] Splicing factors such as heterogeneous nuclear ribonucleoprotein family (hnRNP), serine/arginine-rich splicing factor family (SRSF), heat shock protein family A member 2 (HSPA2) and Y-Box Binding Protein 1 (YBX1) are proteins that facilitate the removal of introns from premature mRNAs in cells.[13] During disease, alterations in the expression and activity of splicing factors lead to changes in IR in mature mRNA.[13]

Nowadays, more and more evidence has demonstrated that aberrant IR patterns have been observed in diverse cancers as well as autoimmune diseases and muscle disorders.[18–20] Several studies have also highlighted the unique role of IR in the activation and proliferation procedures of T cells, B cells and monocytes.[21 22] Moreover, IR has been illustrated as a conserved post-transcriptional regulation mechanism in myocyte development, and specific IR signatures have been found in Duchenne muscular dystrophy, spinal muscular atrophy and Troponin T1-related nemaline myopathy.[23–27] However, the IR profiles have not been systematically studied in IIM subtypes.[28 29]

In this study, we conducted an integrated analysis of IRs, exons and proteomic data in 174 newly diagnosed patients with IIM. We identified IIM-specific differentially expressed IRs (DEIs) and splicing factors. Furthermore, we assessed the impact of DEIs on protein expressions, explored their role in the pathogenesis of IIM and analysed their correlation with clinical parameters and histopathological features.

MATERIALS AND METHODS

Patients with IIM and controls

Patients with IIM were recruited from Xiangya Hospital, Central South University, between June 2016 and March 2023. The cohort for RNA sequencing (RNA-seq) and mass spectrometry (MS) initially comprised 24 controls and 211 IIM (121 DM, 34 IMNM and 56 ASS). In the reverse-transcription (RT)-PCR validation cohort, there were 27 controls, 53 DM, 16 IMNM and 15 ASS. The 2019 European Neuromuscular Centre (ENMC) criteria, 2018 ENMC criteria and Connor criteria were used to determine DM, IMNM and ASS, respectively.[30–32] All patients were newly diagnosed with IIM and had not received glucocorticoid or immunosuppressant treatment. MSAs were tested using Euroline Myositis Profile immunoline-blot (Euroimmun), and samples negative for MSAs or with more than one strongly positive MSA result were excluded. Other exclusion criteria included infection, tumour, severe cardiovascular or metabolic disease, overlap syndrome and sporadic inclusion body myositis (sIBM). Control muscle biopsies were obtained from individuals in a stable phase with osteonecrosis of the humeral head (n=23), osteoarthritis (n=8) or shoulder dislocation (n=20) undergoing shoulder replacement surgery. These subjects had no infections, trauma, tumours, severe cardiovascular or metabolic diseases, autoimmune diseases, muscle weakness, muscle injury or muscular atrophy.

The human experimentation protocol of this study was approved by the ethics committee of Xiangya Hospital (No. 201212074) and conducted anonymously. All patients and controls agreed to participate in the study and provided written informed consent. All procedures involving human participants were performed following the 1964 Helsinki Declaration and its later amendments or comparable ethical standards.

IR and exon identification and analysis

The raw sequencing data were analysed with fastQC (V.0.12.0) and Trimglore (V.0.6.10) to obtain clean reads and then aligned to the reference genome using STAR (V.2.7.10a). For the input of STAR, the genome assembly version used for humans was GRCh38, and the gene model version was ENSEMBL 38.106. Then, the mapped BAM file was used for IR and exon identification and analysis.

For IR identification, our team developed two algorithms to prevent the misclassification of exons as IRs, GTFtools and iREAD, which have been widely employed in numerous studies for the successful identification of IRs in genes.[33–35] The process involves the following steps: (1) We used the standard annotated gene transfer format file obtained from ENSEMBL (ensembl.org/Homo_sapiens/Info/Index, V.106) as a reference. (2) GTFtools exclusively focused on and extracted introns that do not overlap with any exons from any transcript isoforms of any gene. (3) In iREAD, we applied specific criteria for IR identification, including introns with reads ≥20, FPKM (fragments per kilobase of transcript per million mapped reads) ≥3, normalised entropy score ≥0.9 and at least 1 junction read spanning the exon-intron boundary.[36] The raw count matrix of IRs was then input to edgeR (V.3.36.0) to identify DEIs.

For exon identification, we used featureCounts from Subread (V.2.0.3) and subsequently input the raw count matrix of exons into edgeR for the identification of differentially expressed exons (DEEs).

Statistical analysis

Quantitative variables were described as mean (SD) for normally distributed data with homogeneous variances or as median (IQR) for skewed data. Two-group comparisons were conducted using unpaired t-tests for normally distributed data with homogeneous variances, while the Mann-Whitney test was used for skewed data or normally distributed data with non-homogeneous variances. For multiple comparisons, false discovery rate correction was used to adjust the p values. Categorical variables were presented as numbers and analysed using the χ^2 test or Fisher's exact test. Correlations between two continuous variables were analysed using Pearson correlation coefficients (for normally distributed variables) or Spearman's correlation coefficients (for not normally distributed variables). Correlations between continuous variables and categorical variables were assessed using point-biserial correlation or Spearman's correlation coefficients. The normality of all data was evaluated by the Shapiro-Wilks normality test. $P<0.05$ (two-sided) was considered statistically significant. All statistical analyses were performed using SPSS (V.18.0) and R package rstatix (V.0.7.2). ComplexHeatmap (V.2.15.4) and ggplot2 (V.3.4.2) packages of R (V.4.1.2) as well as GraphPad (V.9.5.0) were used for data visualisation.

The detailed methods for clinical feature examination, RNA extraction and sequencing, protein extraction and MS, sample quality control (QC), IR-related splicing factor analysis, real-time PCR and western blot are provided in the supplemental file.

RESULTS

An overview of the general study outline and the analysed process is presented in online supplemental figure 2.

Widespread IR exists in IIM muscle tissues

First, we conducted an IR analysis on 174 muscle samples from IIM individuals (101 DM, 28 IMNM and 45 ASS) as well as 19 controls after QC. A total of 6783 IR events originating from 3111 genes were identified, and the majority of these genes were protein-coding genes (figure 1A). Given that IR is preferentially observed in regions that are high in guanine-cytosine (GC) content and short in length,[37] we categorised the IRs into six groups by length and found that the GC content of each group in the retained introns was significantly higher compared with the non-retained introns (figure 1B). Furthermore, the length of the introns in the recognised retained introns was shorter than that of the non-retained introns (figure 1C). These findings were consistent with previous reports in IRs.[38] We also observed that the identified IRs were widespread across chromosomes (figure 1D).

Characterised IR profiles and splicing factors in IIM main subtypes

Principal component analysis showed a distinct separation of IR profiles between IIM main subtypes and the controls (figure 2A). Compared with the control group, DM, IMNM and ASS groups exhibited upregulation of 1289, 961 and 864 DEIs, respectively, while downregulation was observed in 1079, 974 and 797 DEIs, respectively (figure 2B, online supplemental data 1–3). To further investigate the differences between IIM subtypes, we selected the top 30 highly elevated DEIs for clustering analysis, which effectively distinguished DM, IMNM and ASS groups from the control group (figure 2C). Notably, the ASS muscles exhibited moderately high expression of IMNM top genes as well. Among the DEIs, we found 649, 508 and 144 IRs specifically dysregulated in DM, IMNM and ASS muscles, respectively (figure 2D). The genes exhibiting specific IRs in DM, IMNM or ASS also demonstrated a certain degree of specificity at the exon and protein levels (online supplemental figure 3).

Functional enrichment analysis of the IIM subtype-specific DEIs highlighted notable pathways affected in different subtypes (figure 2E). In the DM group, there was involvement in the cytokine-mediated signalling pathway, IκB kinase/nuclear factor (NF)-κB signalling pathway, type 1 IFN pathway, endothelial cell morphogenesis and interleukin (IL)-13 production. For IMNM, findings included muscle contraction and regeneration, myotube differentiation, myoblast fusion, skeletal muscle satellite cell proliferation, tumour necrosis factor (TNF) response and IL-8 production. In ASS, enriched pathways encompassed metabolic processes such as cellular carbohydrate metabolic process, glycogen metabolic process, energy reserve metabolic process, polysaccharide metabolic process, monocarboxylic acid biosynthetic process, phospholipid biosynthetic process, bile acid metabolic process, and nitric oxide metabolic process.

We also identified 1037 shared DEIs across various IIM subgroups. Functional analysis revealed that these DEIs were enriched in biological processes related to muscle cell differentiation, protein translation, mRNA stabilisation and response to endoplasmic reticulum stress (ERS) (online supplemental figure 4A). Within the shared DEIs, 536 increased IRs in controls were enriched in processes related to lipid metabolism, RNA splicing, and differentiation in muscle cells, T cells, myeloid cells, endothelial cells, and fat cells (online supplemental figure 4B).

As the alteration of IR profiles is mainly regulated by splicing factors, we further investigated the regulation networks of splicing factors in IIM subtypes by SPIRON. A total of 192 splicing factors in the U1-type major splicing pathway, the U12-type minor or atypical splicing pathway and spliceosome were collected as reported previously.[36] The correlation between the expression of each IR and 192 splicing factors was evaluated. In the control group, we observed a wide range of splicing factors associated with the IRs, such as RMB5, HSPA2 and SRSF7 (figure 2F). In DM, splicing factors such as YBX1, PCBP2 and SNRPD3 showed strong correlations with the identified IRs, particularly YBX1 (figure 2F). Similarly, in IMNM, HSPA2 appeared to be the most significant splicing factor. In ASS, splicing factors YBX1, COLEC12, SNRNP48 and SNRPD3 exhibited robust connections with IRs; however, none of them were in a dominant position. The protein levels of YBX1 and HSPA2 were also found to be higher in DM or IMNM compared with controls (figure 2G).

The influence of IR on protein expression in IIM muscles

To evaluate the impact of IR on protein expression, we scrutinised the protein expression of 762 genes with DEIs by using data from 193 samples with aligned RNA-seq and MS data. Log2-transformed fold changes (logFC) were used for IR and protein expression between IIM and control samples, denoted as logFC(IR) and logFC(protein), respectively. Our results revealed that the protein expression of genes with higher IR tends to decrease more than that of the genes with lower IR (figure 3A). For example, when IR expression is elevated in IIM subtypes (ie, $\mathrm{logFC(IR)}>0$), the logFC(protein) value is mostly smaller than logFC(IR) (ie, below the diagonal line). Conversely, when IR expression is reduced in IIM subtypes (ie, $\mathrm{logFC(IR)}<0$), the logFC(protein) value is predominantly higher than logFC(IR) (ie, above the diagonal line). This observation implies that protein expression tends to decrease when mRNA exhibits a higher level of IR.

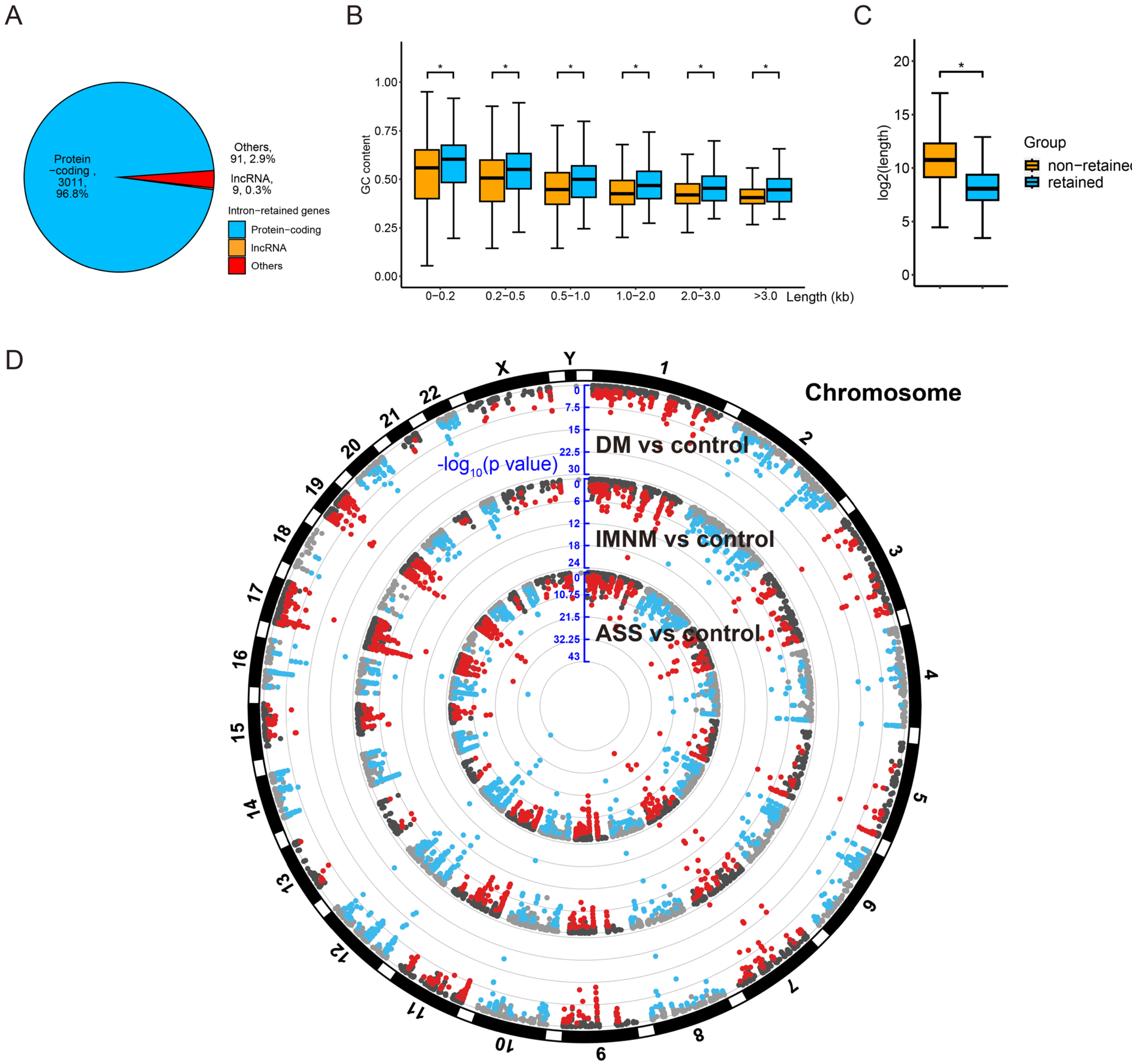

Figure 1 Intron retention (IR) events in idiopathic inflammatory myopathy muscles. (A) Distribution of intron-retained genes by subtype. (B) Guanine-cytosine (GC) content comparison between retained and non-retained introns binned by length (p value: *, <0.05). (C) Comparison of length between retained and non-retained introns (p value: *, <0.05). (D) Distribution of IR events in dermatomyositis (DM), immune-mediated necrotising myopathy (IMNM) and antisynthetase syndrome (ASS) compared with control muscles across chromosomes. Highlighted IRs had p<0.05. ASS, antisynthetase syndrome; DM, dermatomyositis; GC, guanine-cytosine; IMNM, immune-mediated necrotising myopathy.

In most situations, at the single gene level, if two genes have the same exon expression, the one that has a higher IR would have lower protein levels. Therefore, we investigated the relationship between IR/exon ratio and protein abundance. We found that the IR/exon ratio was negatively correlated with protein abundance in both controls and IIM subtypes (figure 3B).

Among the dysregulated proteins in IIM subtypes, 12.7%–20.9% (12.7% in DM, 20.9% in IMNM and 16.9% in ASS) of the changed proteins exhibited only IR alterations, while 15.6%–20.0% (20.0% in DM, 15.6% in IMNM and 15.8% in ASS) showed only exon alterations (online supplemental figure 5). Additionally, 25.9%–36.3% (36.3% in DM, 31.0% in IMNM and 25.9% in ASS) of changed proteins exhibited alterations in both exon and IR (online supplemental figure 5).

We also examined the impact of IR locations on exons and proteins. The results indicated that IRs situated in the 5' UTR, 3' UTR or coding sequence regions have a comparable effect on exon expressions (online supplemental figure 6A–C left panel). However, in DM and IMNM, IRs located in the 5' UTR region had a more significant impact on protein expression compared with those in the 3' UTR or coding sequence region (online supplemental figure 6A–C right panel). Additionally, in ASS, when the protein was upregulated, IRs located in the 5' UTR region had a more pronounced effect on the upregulation of the protein (online supplemental figure 6A–C right panel), indicating that the location of IRs influences protein expression.

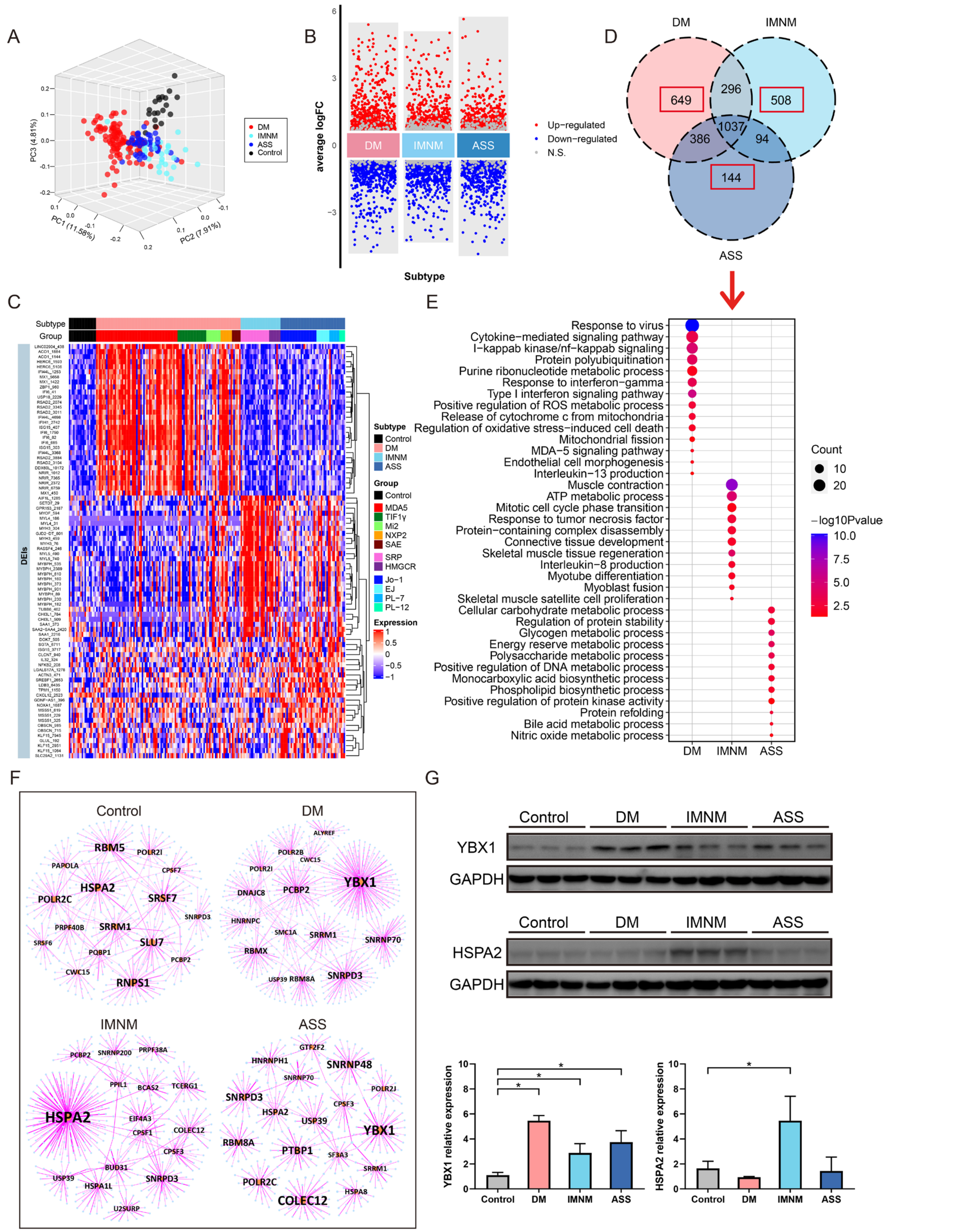

Figure 2 Differentially expressed retained introns (DEIs) and splicing factors in idiopathic inflammatory myopathy (IIM) subtypes. (A) Principal component analysis plot showing the IR profiles in dermatomyositis (DM), immune-mediated necrotising myopathy (IMNM), antisynthetase syndrome (ASS) and control muscles. (B) DEIs in DM, IMNM and ASS compared with the control group. DEIs with p<0.05 were highlighted. (C) Heatmap showed the 30 most upregulated DEIs in IIM subtypes. (D) Venn diagram showed the overlapped DEIs among IIM subtypes. (E) Gene ontology (GO) biological process terms enriched in specific intron-retained genes of DM, IMNM, and ASS. The colours of the dots represent p values, and the size of the dots represents the number of IR-related genes associated with the GO term. (F) The regulatory network between IRs and splicing factors in DM, IMNM, Ass and control muscles. Lines represent connections between IRs and splicing factors, while orange and blue circles represent splicing factors and IRs, respectively. Splicing factors with more than 20 first-degree neighbours are labelled. The thickness of the lines represents the weight of IR to its splicing factor. (G) The protein expression of YBX1 and HSPA2 in IIM subtypes. ASS, antisynthetase syndrome; DM, dermatomyositis; IMNM, immune-mediated necrotising myopathy.

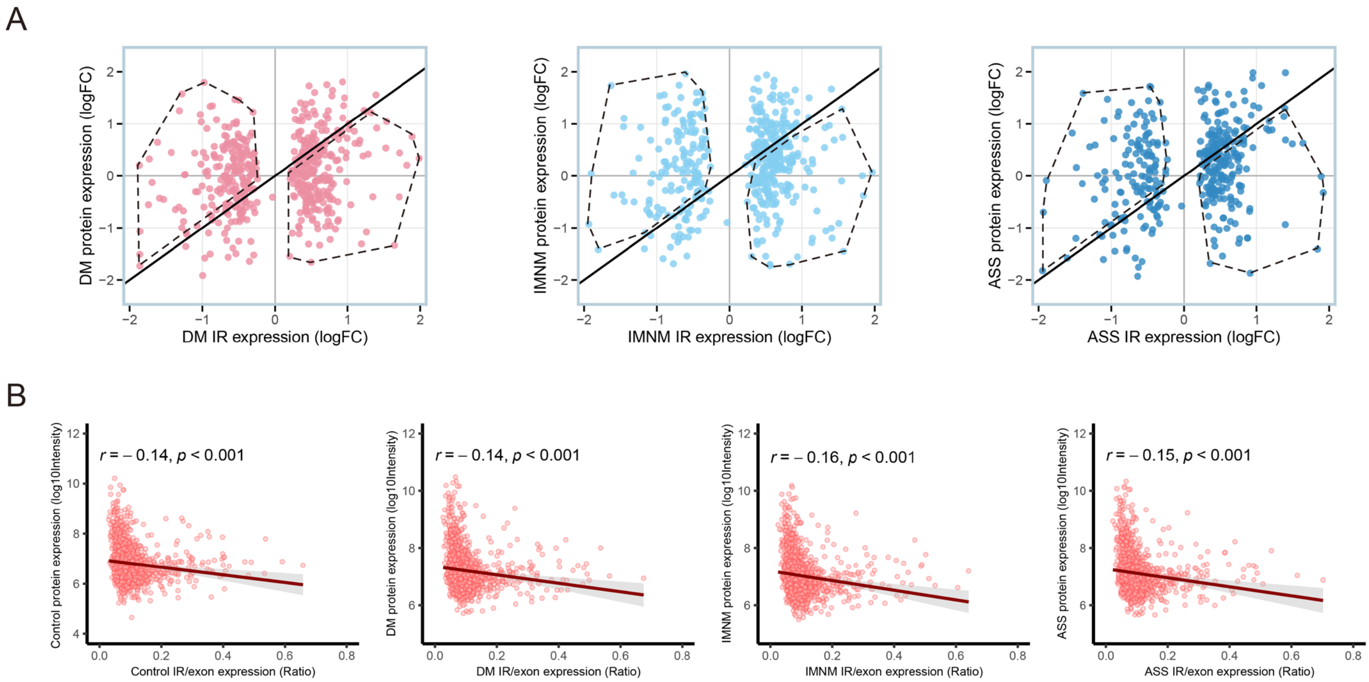

Figure 3 The role of intron retention (IR) in regulating protein abundance in idiopathic inflammatory myopathy (IIM) muscles. (A) Increased IR expression is associated with a reduced level of protein expression. The X-axis represents the log2-transformed fold change (logFC) of IR expression between IIM subtypes and controls, while the Y-axis represents the log2-transformed FC of protein expression between IIM subtypes and controls. Each point corresponds to a single gene. For the majority of genes, increased IR expression is associated with a reduced level of protein expression in IIM subtypes. When the IR expression is higher in dermatomyositis (DM), immune-mediated necrotising myopathy (IMNM) or antisynthetase syndrome (ASS) (ie, IR expression logFC>0), the protein expression value is mostly smaller than the IR expression (ie, below the diagonal line, highlighted with a dashed line). Conversely, when the IR expression is lower in DM, IMNM or ASS (ie, IR expression logFC<0), the protein expression value is mostly higher than the IR expression (ie, above the diagonal line, highlighted with a dashed line). (B) The correlation between IR/exon ratio and protein expression in IIM subtypes and controls. ASS, antisynthetase syndrome; DM, dermatomyositis; IMNM, immune-mediated necrotising myopathy; IR, intron retention; logFC, log2-transformed fold change.

Specific IRs in DM muscles

To further explore the roles of specific IRs in DM muscles, we conducted an integrated analysis of the profiles of DEIs, DEEs and differentially expressed proteins in DM biopsies compared with controls. A total of 21 genes exhibited upregulation at the IR level and downregulation at the protein level, with no significant changes or upregulation at the exon level (online supplemental figure 7A,B). Meanwhile, 18 genes showed decreased expression levels at the IR level and increased expression levels at the protein level with no significant changes or an increase at the exon level. These genes displayed a distinct expression pattern where high levels of IR were associated with low protein expression, while low levels of IR were linked to high protein expression, irrespective of exon levels. Among the 39 IR-related genes, 24 were specifically identified in DM, and 7 genes were enriched in DM-related pathways (figure 4A, online supplemental table 5). In the IFN-related genes, CDC34 showed elevated expression at the exon and IR levels but decreased expression at the protein level. Gelsolin (GSN) displayed decreased expression at both exon and IR levels but increased expression at the protein level. Among the cytokine-related genes, SERPINB1, KLF4 and NCL showed decreased IR expression but elevated protein expression. Another cytokine-related gene TLE5 and the NF-κB-related gene PRDX1 displayed high expression at both exon and IR levels but low expression at the protein level.

Additionally, the IR levels of these genes were examined in the muscles of DM patients with anti-MDA5+, anti-TIF1γ+, anti-Mi2+, anti-NXP2+ and anti-SAE+ antibodies, compared with the control group, and the majority of the genes were dysregulated in multiple antibody-defined groups (figure 4B).

In addition to CDC34 and GSN, we evaluated the IFN-related genes that were simultaneously detected at the IR, exon and protein levels. The expression of many IFN-related genes at the protein and IR levels exhibited contrasting trends, such as DXH58 and BST2, even though the gene expression at the exon level may show significant changes (figure 4C). Although MX1, MX2 and IFIH1 were upregulated at the exon, IR and protein levels, the fold change of protein alteration was significantly lower compared with that of the exons. We also validated our findings in IRs via RT-PCR. The IRs of CDC34 were significantly higher in DM, while the IRs of SERPINB1 and KLF4 were significantly lower in DM compared with the control, although the exons showed comparable or similar trends as the IRs (figure 4D). Consistent with the RT-PCR results, the coverage plots of the RNA-seq data confirmed the altered IRs in these genes in DM (figure 4E).

Specific IRs in IMNM muscles

In IMNM, 34 genes with opposing trends in IR and protein expression were identified, with 17 of them being specific to IMNM (online supplemental figure 7C,D). Among these genes, 7 genes enriched in muscle regeneration, muscle contraction process and TNF response were selected for further analysis (online supplemental table 6). The expression of ANXA1, CMYA5 and NOL3 exhibited a decline at the IR level, an increase at the protein level and no significant difference at exon

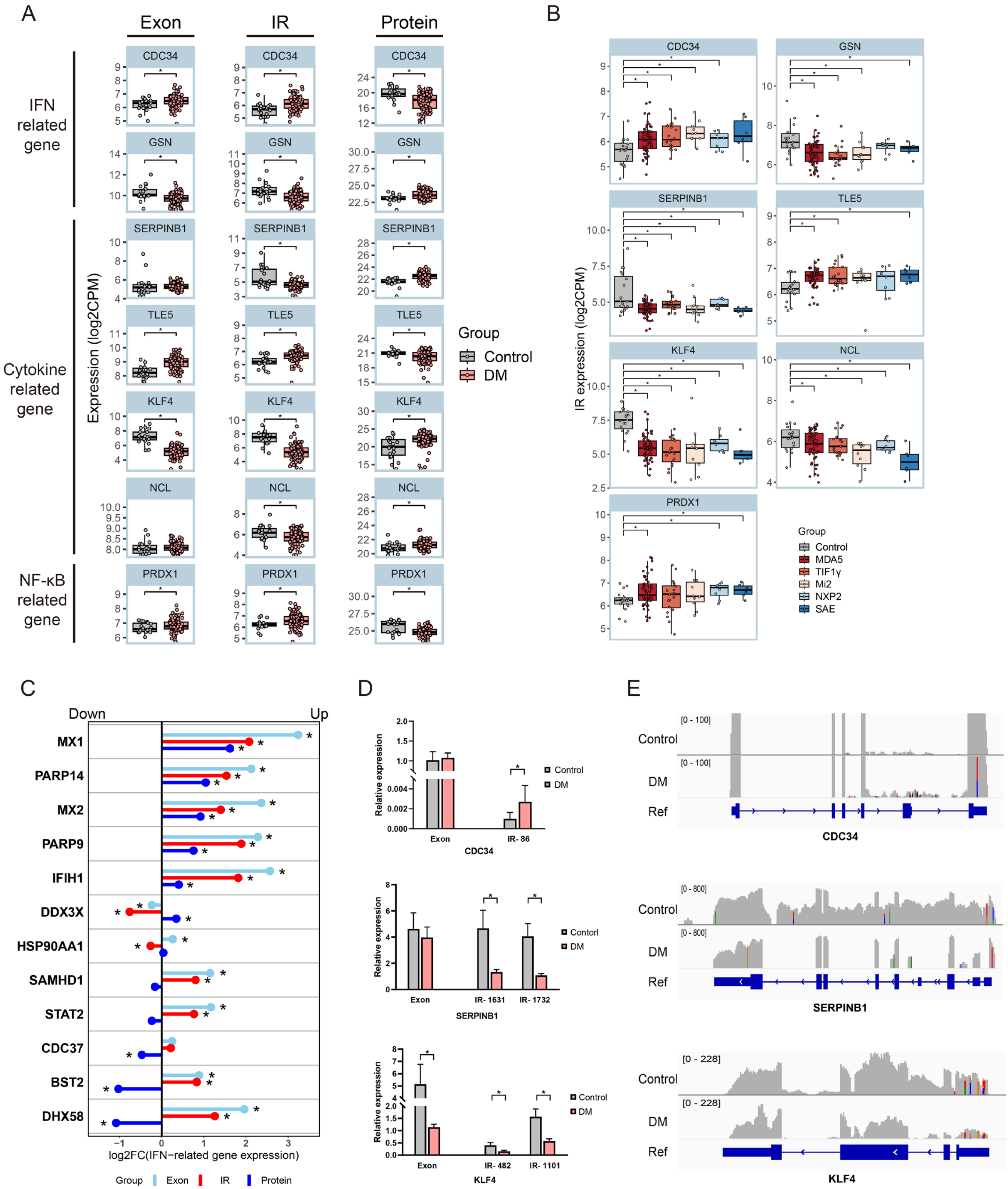

Figure 4 Intron retention (IR) profiles in dermatomyositis (DM) muscles. (A) The expressions of IRs, exons and proteins in genes associated with DM pathogenesis. (B) The expression of IRs in DM subtypes defined by myositis-specific autoantibodies. (C) Expression of interferon-related genes at the exon, IR and protein levels in DM muscles. (D) Reverse-transcription-PCR validation of selected genes at exon and IR levels between DM and control muscles. (E) Coverage plots of selected genes in DM compared with control in RNA sequencing. One representative from each group was shown (p value: *, <0.05). DM, dermatomyositis; IFN, interferon; IR, intron retention.

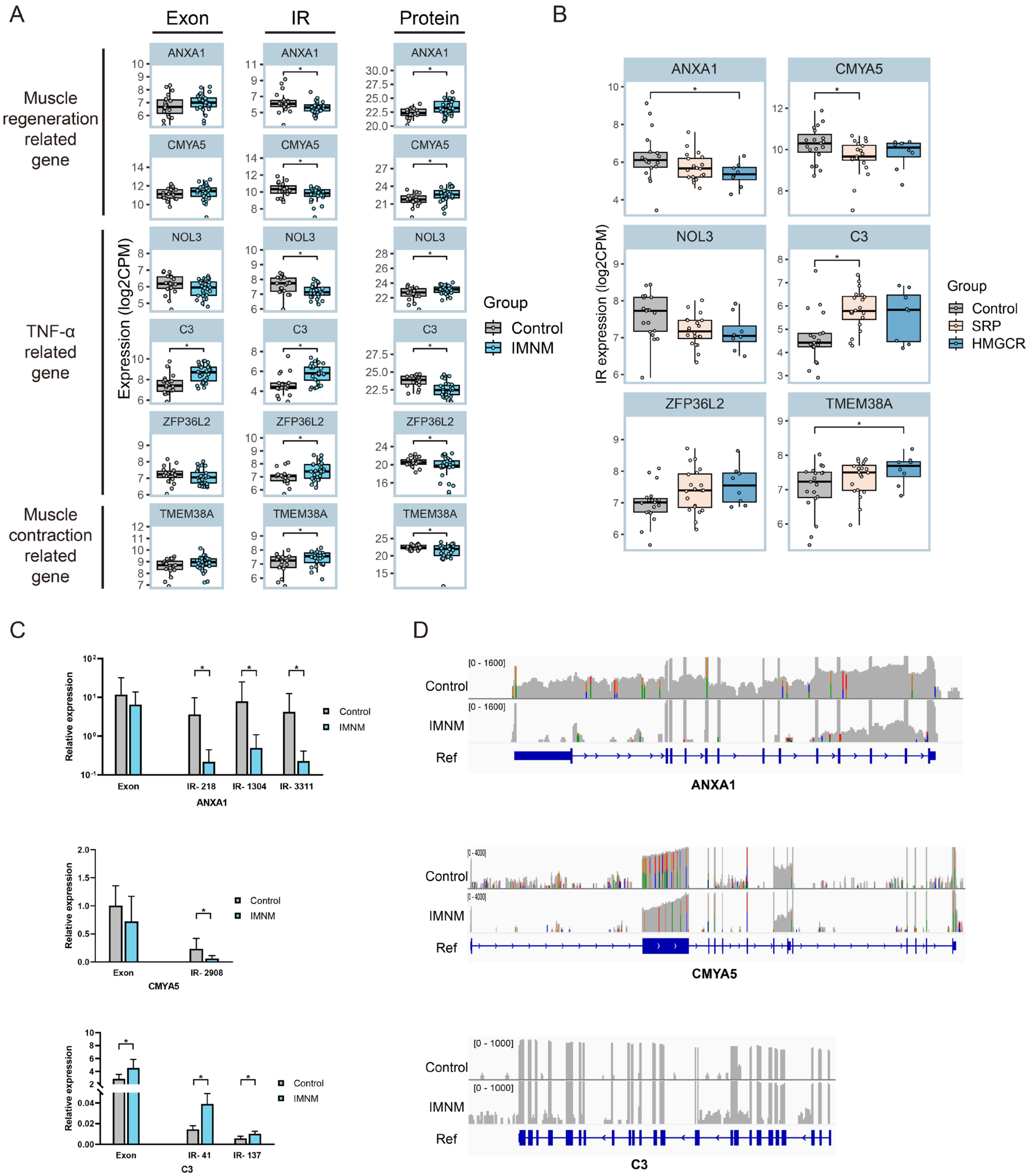

Figure 5 Intron retention (IR) profiles in immune-mediated necrotising myopathy (IMNM) muscles. (A) The expressions of IRs, exons and proteins in genes associated with IMNM pathogenesis. (B) The expression of IRs in IMNM subtypes defined by myositis-specific autoantibodies. (C) Reverse-transcription-PCR validation of selected genes at exon and IR levels between IMNM and control muscles. (D) Coverage plots of selected genes in IMNM compared with control in RNA sequencing. One representative from each group was shown (p value: *, <0.05). HMGCR, 3-hydroxy-3-methylglutaryl-CoA reductase; IMNM, immune-mediated necrotising myopathy; IR, intron retention; TNF-α, tumour necrosis factor alpha.

levels (figure 5A). However, the expression of C3, ZFP36L2 and TMEM38A was upregulated at the IR level and downregulated at the protein level. Additionally, specific IMNM subtypes defined by MSAs may exhibit distinct phenotypes of IR expression patterns. The IR expression of CMYA5 was significantly lower, and C3 was significantly higher in patients with anti-SRP+ compared with controls (figure 5B). Furthermore, the IR expression of ANXA1 was significantly decreased, whereas TMEM38A showed significant increases in patients with anti-HMGCR+ compared with controls (figure 5B). The mRNA levels of ANXA1, CMYA5 and C3 at different IR sites were verified by RT-PCR (figure 5C), and the coverage plots of ANXA1, CMYA5 and C3 genes displayed consistent results with RT-PCR findings (figure 5D).

Specific IRs in ASS muscles

15 genes with high IR and low protein expression, as well as 8 genes with low IR and high protein expression, were identified in ASS (online supplemental figure 7E,F). Among them, 15 genes were specifically associated with ASS. Considering IR-related genes were enriched in metabolism pathways in ASS muscles, we included eight identified genes that were involved in carbohydrate, protein, lipid, DNA and ATP metabolism processes (online supplemental table 7). Notably, ITPKA, PKN1, PHB2, ECI2 and ATP5F1D exhibited elevated IR levels and decreased protein levels (figure 6A). CSPG4, VPS35 and HNRNPA2B1 showed increased IR levels and declining protein levels. Specific regulation patterns were observed among the MSA-defined subtypes of ASS, such as downregulated VPS35 IRs in patients with anti-Jo-1+, as well as upregulated PHB2 and ATPF1D in patients with anti-Jo1+ (figure 6B). The IR level of HNRNPA2B1 was altered in multiple MSA-defined groups in ASS. Additionally, we performed RT-PCR to verify the exon and IR expression and examined the read coverage of genes such as ECI2, HNRNPA2B1 and ATP5F1D (figure 6C,D).

Correlations between IIM subtype-specific IRs and clinical parameters

To assess the potential clinical significance of IR in IIM, we conducted a correlation analysis between IIM subtype-specific IRs and clinical manifestations, laboratory findings as well as histopathological features (figure 7A). Specific IRs for DM, such as GSN and SERPINB1, exhibited a negative correlation with skin rash (heliotrope rash and Gottron's papules) and MMT8 score but were positively correlated with muscle enzymes (alanine transaminase, aspartate aminotransferase, lactate dehydrogenase, CK, myoglobin), as well as myofiber necrosis and regeneration, while TLE5 exhibited the opposite trend. Additionally, PRDX1 displayed a negative correlation with COX-negative fibres. Notably, all IMNM-specific IRs (CMYA5, NOL3, TMEM38A, ANXA1, C3 and ZFP36L2) were strongly correlated with MMT-8 score and muscle enzymes, as well as myofiber necrosis and regeneration; some of them were also correlated with skin rash and muscle inflammation. Regarding ASS-specific IRs, ITPKA and CSPG4 showed negative correlations with ILD and MMT8 score but had positive associations with muscle enzymes, myofiber necrosis, regeneration and membrane attack complex.

We also identified a distinctive pattern of IR expressions that gradually changed across the histopathological features of the none, mild and high groups, including necrosis, regeneration, CD4+ T cell infiltration, CD8+ T cell infiltration and CD68+ macrophage infiltration (figure 7B). During the progression of necrosis and regeneration, the IRs of ISGs, such as IFI44, IFIH1, IFN44L, ISG15 and MX1, progressively decreased from none to high groups, while the IR levels of IGHG1, SAA1, MYL4 and CXCL12 gradually increased. IRs in immune response-associated genes, such as HLA-C, GRN, IGHM and TRIM14, showed a gradual elevation during inflammation development. Furthermore, the IR levels of ROS1, CXCL2 and FOS decreased, while NKG7, LGALS9 and RUNX1 increased during CD4+ T cell infiltration. Regarding CD8+ T cell infiltration, FYB2, MYLK4 and FAM166B were downregulated, while HLA-B, RUNX1 and HLA-A were upregulated. In the case of CD68+ macrophage infiltration, JAK1, ISG15 and MX1 decreased, while CSF1R, IL6 and CD14 increased.

As IFIH1 encodes the MDA5 protein, TRIM21 encodes the Ro-52 protein, and SRP14 encodes the SRP protein. To examine IR levels in IFIH1, TRIM21, and SRP14 concerning antibody levels, correlation analyses were conducted. Figure 7C illustrates strongly positive correlations between IR levels of IFIH1 and TRIM21 and anti-MDA5 antibodies, while IR levels of SRP14 strongly correlated with anti-SRP antibodies.

To further explore the potential connection between IRs and ILD, we categorised patients with IIM into ILD and no-ILD groups. In the ILD group, the IRs of NOL3, ATP5F1D and TMEM38A were upregulated, while the IRs of ANXA1, ITPKA, APOE and XIRP1 were downregulated (online supplemental figure 8A). Functional enrichment analysis of the altered IRs revealed significant changes in processes such as the generation of precursor metabolites and energy, regulation of peptidase activity, response to oxidative stress, extracellular matrix organisation, response to TGF-β and other metabolic processes in muscles when ILD occurs in patients with IIM (online supplemental figure 8B).

DISCUSSION

IRs play crucial roles in the regulation of gene expression and biological processes, contributing to the pathogenesis of autoimmune diseases.[29 39 40] In this study, we conducted a comprehensive investigation of IR profiles and evaluated their roles in the pathogenesis of different IIM subtypes. Notably, we observed a counterintuitive disparity between protein and exon/intron expression, exemplified by genes such as CDH5, STAT2 and IGHG1, among others. Similar observations have been reported in various published datasets.[41–43] In many situations, the expression patterns at the exon and protein levels may not align, highlighting the complexity of gene regulation.[43] More than 60% of the variation in protein concentration cannot be explained by measuring mRNA abundance alone.[44] Post-transcriptional regulation, encompassing mRNA splicing and mRNA polyadenylation, represents crucial mechanisms in the regulation of protein expressions, contributing to the divergent trends observed between proteins and exons.[44] In the context of alternative splicing, IRs in mRNA emerge as a significant mechanism capable of affecting the stability, distribution or translational efficiency of transcripts.[45]

We revealed that protein expression tends to decrease when the mRNA contains higher levels of IR expression. The IR/exon ratio exhibited a negative correlation with protein abundance, suggesting a potential negative regulatory role of IRs in the protein expression of IIM muscle tissue. The regulatory impact of IRs on protein expression in transcripts could vary in multiple ways. For instance, when mRNA is retained in the nucleus with high IR, it may undergo nuclear degradation by exosomes or remain stable until a signal for further splicing is received.[46 47] Cytoplasmic mRNA with high IR in the coding sequence can be targeted by the NMD machinery for degradation.[48] Moreover, we observed that IRs located in the 5' UTR region of transcripts had a more pronounced influence on protein expression compared with those located in the 3' UTR or coding sequence region in DM and IMNM.

Our study identified IIM subtype-specific IRs, showing strong enrichment in pathways associated with the pathogenesis of

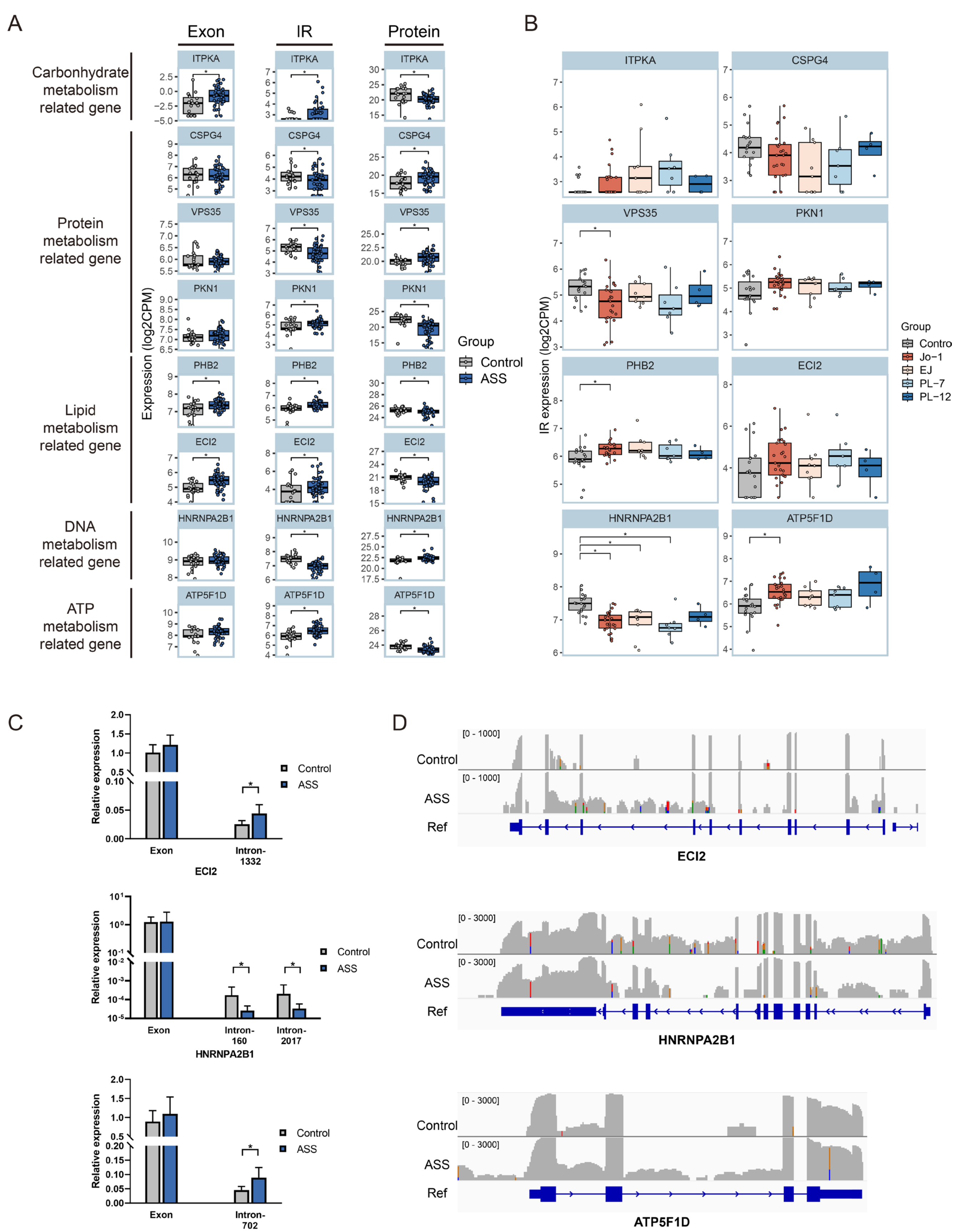

Figure 6 Intron retention (IR) profiles in antisynthetase syndrome (ASS) muscles. (A) The expressions of IRs, exons and proteins in genes associated with ASS pathogenesis. (B) The expression of IRs in ASS subtypes defined by myositis-specific autoantibodies. (C) Reverse-transcription-PCR validation of selected genes at exon and IR levels between ASS and control muscles. (D) Coverage plots of selected genes in ASS compared with control in RNA sequencing. One representative from each group was shown (p value: *, <0.05). ASS, antisynthetase syndrome; IR, intron retention.

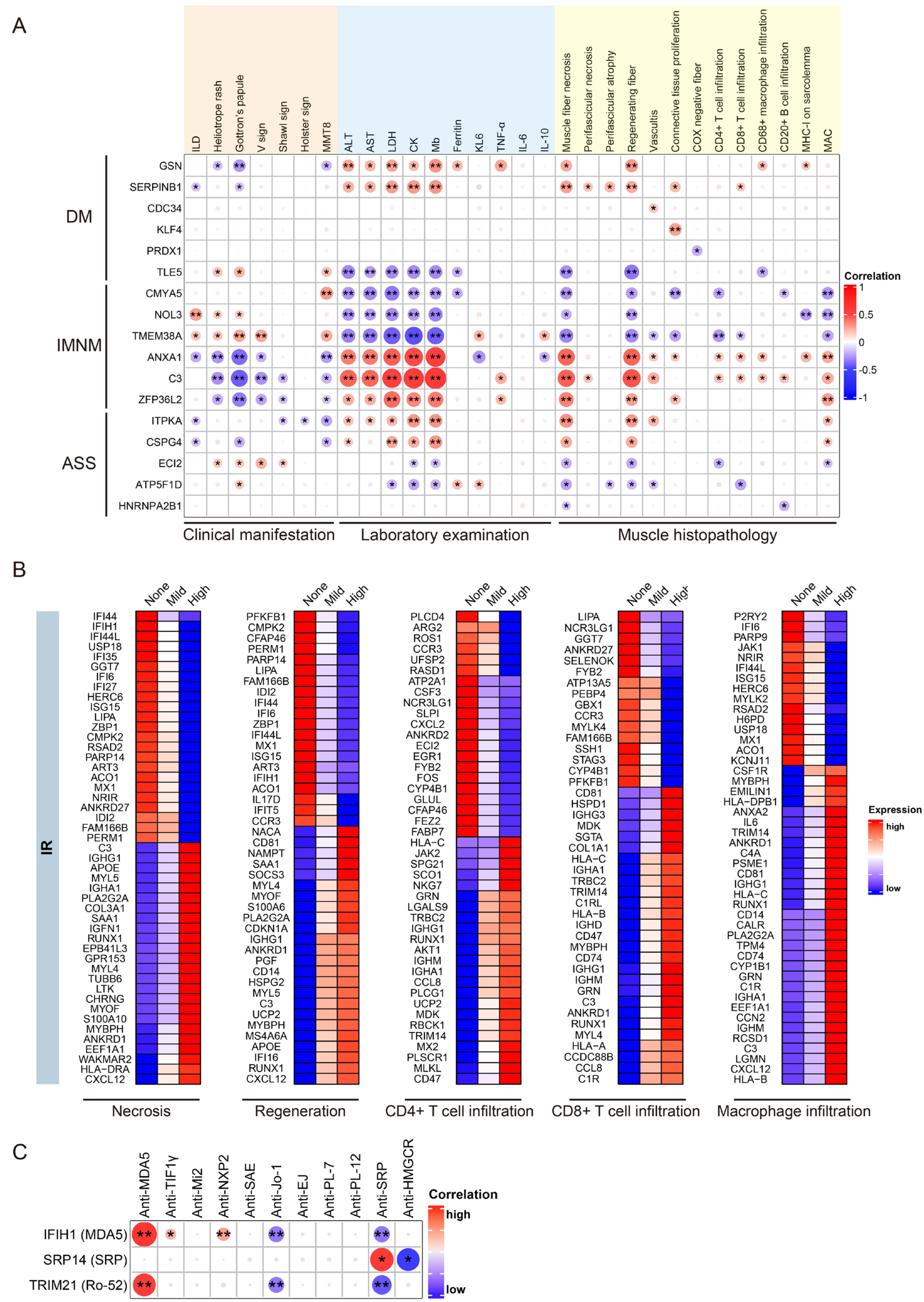

Figure 7 Correlation between differentially expressed IRs (DEIs) and clinical parameters. (A) Correlation between DEIs and clinical manifestations, laboratory examinations and muscle histopathology. The colours and size of the dots represent the correlation coefficient. (B) IR profiles in the none, mild and high groups of necrosis, regeneration, CD4+ T cell infiltration, CD8+ T cell infiltration and CD68+ macrophage infiltration. (C) The correlation between autoantibody levels and the IR expression in antigens of anti-MDA5, anti-SRP and anti-Ro-52 antibodies. The colours and size of the dots represent the correlation coefficient (p value: *, <0.05; **, <0.01). ASS, antisynthetase syndrome; DM, dermatomyositis; IMNM, immune-mediated necrotising myopathy; IR, intron retention.

respective IIM subtypes. For example, IFN-related pathways were prominent in DM, while skeletal muscle cell differentiation and proliferation processes were enriched in IMNM. Moreover, we found that the DM-specific DEIs were also involved in endothelial cell morphogenesis and IL-13 production, while IMNM-specific DEIs participated in TNF response and IL-8 production. ASS-specific DEIs were involved in some special metabolic processes such as monocarboxylic acid biosynthetic processes, bile acid and nitric oxide metabolic processes. Among the IIM subtype-specific IRs, SERPINB1 showed low IR levels and high protein levels in DM, while ANXA1 exhibited low IR levels and high protein levels in IMNM. SERPINB1 possesses the ability to prevent programmed necrosis and cytokine release of neutrophils and monocytes, while ANXA1 can facilitate leucocyte emigration into inflamed tissue, promote the differentiation of myeloid cells and enable T cell differentiation into proinflammatory Th1 cells.[49–52] These results indicate that the presence of IRs in genes may have an impact on protein expression in pathways that contribute to the development of IIM subtypes.

In addition to subtype-specific DEIs in IIM, we identified commonly altered IRs across IIM subgroups, showing enrichment in cell development and the regulation of mRNA, protein synthesis and response to ERS. ERS encompassing both the ER overload response and the unfolded protein response (UPR) plays a role in the molecular underpinnings of skeletal muscle damage across all IIM subtypes.[53 54] Overexpression of major histocompatibility complex class I (MHC-I) and the accumulation of unfolded or misfolded proteins in myofibers can trigger an ERS response, activating NF-κB signalling and initiating the UPR pathway. This cascade leads to the production of myokines, cytokines or cell death.[53 54] Additionally, dysregulated IRs in genes involved in 3' UTR-mediated mRNA stabilisation may also play a role in the pathogenesis of IIM. Thus, these results suggest that these pathways were commonly affected in DM, IMNM and ASS muscles. Among these shared DEIs, 536 DEIs exhibit higher expression in control muscles, mainly involved in muscle tissue development, RNA splicing and cell differentiation processes. The IR in control muscles may be preserved in the nucleus or cytoplasm, serving as a reservoir and waiting for a signal to activate further 'delayed' splicing and translation into proteins.[55] These findings align with previous reports suggesting that certain genes in inactive cells can exhibit high levels of IRs, as seen in MYOD1 in muscle stem cells, CASP3, COPS3, PSMB6 and PSMD8 in CD4+ T cells and CXCL2 and NFKBIZ in macrophages.[48 56 57] On activation, the IRs in these cells could be significantly reduced.[30 31 33]

Some IIM subtype-specific IRs are closely associated with skin rash, muscle enzyme levels and histopathological features. Additionally, a unique IR pattern was observed in muscles displaying necrosis, regeneration or inflammation cell infiltration. The IRs of ISGs, such as IFIH1 and MX1, exhibited a progressive downregulation in the none, mild, and high groups of necrosis and regeneration, while the IRs of cytokines CCL8 and IL6 demonstrated a gradual increase in muscle T cell or macrophage infiltration. These IRs could actively contribute to the pathogenesis of IIM, resulting in muscle necrosis or inflammation. Further investigation is required to explore the potential relationship between changes in intron expression and fibre necrosis or inflammation.

MSAs have been recognised as a hallmark of IIM. The existence of anti-SRP, anti-HMGCR and anti-TIF1γ antibodies in muscle tissue has been reported to mitigate myofiber damage, including degeneration, atrophy and necrosis.[58 59] However, the mechanisms underlying MSA production and the specific target organ damage remain unknown. It has also been reported that transcripts with IRs, regardless of whether they are translated or degraded, could generate peptides for endogenous processing, proteolysis cleavage and presentation on MHC-I.[60] The presented peptides may activate immune cells and serve as antigens for the generation of autoantibodies in muscles or attract immune cell infiltration.[61 62] We identified that the IRs of IFIH1 (encoding MDA5) and TRIM21 (encoding Ro-52) exhibited a positive correlation with the levels of anti-MDA5 antibodies, while the IRs of SRP14 showed a positive correlation with the levels of anti-SRP antibodies. Further studies are needed to explore whether these IRs can affect the production of anti-MDA5 and anti-SRP antibodies. Moreover, it has been reported that IRs in certain genes (CXCL2, DNAJB1 and RIOK3) can be induced or correlated with IFNs,[48 63] indicating that IFN also plays a crucial role in IR. As IFIH1 is an IFN-inducible gene, additional research is required to determine whether IFN also regulates IRs in IFIH1, induces abnormal MDA5 antigens and subsequently triggers MDA5 antibodies.

Splicing factors play a key role in regulating IR events, and many of them are known to contribute to disease progression.[64] Among these factors, HSPA2 and YBX1 have already been recognised as biomarkers in inflammation, and they are considered potential therapeutic targets in myositis or myopathy.[65 66] HSPA2 is a spliceosome component abundantly expressed in muscles and is implicated in a wide range of cellular processes, including the folding of newly synthesised polypeptides, proteolysis of misfolded proteins and the formation of protein complexes.[67 68] It was upregulated in the muscle and serum of patients with DM and participated in UPR during protein degradation in myofibers of MHC-I overexpression-induced myopathy in mice.[69 70] YBX1, a cold-shock protein, belongs to the major splicing pathway and is involved in various cellular processes, such as translational repression, RNA stabilisation and mRNA splicing.[71] It was elevated in proliferative myoblasts and myotubes during fusion procedures in vivo.[72] In our study, we identified HSPA2 and YBX1 as the most crucial splicing factors that undergo significant changes in the muscles of patients with DM and IMNM, leading to alterations in IR. Considering MHC-I overexpression and UPR leading to ER stress are also important features of DM muscles, while myofiber regeneration and myoblast proliferation are main characters of IMNM muscle,[73] targeting these aberrant splicing factors may hold potential for the development of novel therapeutic approaches for the treatment of IIM.

This study has some limitations: (1) The number of control muscles was limited, and they were not from the healthy control group. Future studies would require a larger sample size of control specimens. (2) Although we demonstrated that IR was widespread in IIM muscles and correlated with clinical manifestations, the changes in IR in response to treatment are needed for further study. (3) Patients with sIBM were excluded from the study due to it being rarely seen in Chinese patients with IIM. (4) Functional validation and investigation into the exact mechanisms of the identified IRs and splicing factors need to be conducted in animal models of IIM.

CONCLUSION

Our research has identified disturbed IR profiles in the muscles of IIM subtypes. We also discovered potential dysregulated splicing factors in DM, IMNM and ASS that were associated with these IR alterations. Through the integration of RNA-seq and MS data, we found that IR could potentially influence protein expression in IIM muscles and revealed possible

IR signatures that are closely related to clinical features. These novel IR and splicing factor biomarkers we have identified may provide valuable insights into the understanding of pathogenesis, enable early classification diagnosis and serve as promising therapeutic targets for IIM in the future.

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Acknowledgements We thank all the rheumatologists in the Department of Rheumatology and Immunology at Xiangya Hospital, Central South University, for their invaluable support and clinical assistance throughout this research. We also thank all patients and volunteers who participated in this study and generously provided us with biopsy samples.

Contributors HZ, YL and HL designed the study. YX, YL, HZ and XZ participated in data and sample collection. YL and YX acquired the data. YX, SX, H-DL and HZ analysed and interpreted the data. YX and HZ wrote the final manuscript. All authors approved the final manuscript. HZ is the guarantor of this study.

Funding This study was supported by the National Natural Science Foundation of China (82202002) and the Natural Science Foundation of Hunan Province, China (2022JJ30993, 2023JJ41002).

Competing interests None declared.

Patient and public involvement Patients and/or the public were not involved in the design, or conduct, or reporting, or dissemination plans of this research.

Patient consent for publication Not applicable.

Ethics approval This study involves human participants. The human experimentation protocol of this study was approved by the ethics committee of Xiangya Hospital, Central South University (No. 201212074). Participants gave informed consent to participate in the study before taking part.

Provenance and peer review Not commissioned; externally peer reviewed.

Data availability statement Data are available upon reasonable request. All data are available in the main text or the supplementary materials. The raw data from RNA sequencing and MS analyses are deposited in the institutional database and will be provided upon request to the corresponding author.

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Supplementary Material

Title: Characterized intron retention profiles in muscle tissue of idiopathic inflammatory myopathy subtypes

Supplementary Materials and Methods

Clinical manifestation, laboratory examination and muscle histopathology

For all participants, clinical data, including demographic characteristics, clinical manifestations, laboratory findings, and histopathology features, were recorded (supplementary table 1, 2, 3). The examination of muscle weakness was conducted using the MMT8 score, involving testing neck flexors, deltoid, biceps, gluteus maximus, gluteusmedius, quadriceps, wrist extensors, and ankle dorsiflexors within a range of 0-80.[1]

Muscle biopsies were obtained from the biceps brachii muscle of IIM patients. To match the biopsy site of IIM patients, control muscles were also collected from the biceps brachii. The muscle samples were immediately embedded in gum tragacanth, immersed with pre-cooled isopentane by liquid nitrogen for histopathological examination, or directly immersed and stored in liquid nitrogen for RNA sequencing and MS analysis. Histopathological features of the biopsies were independently analyzed by two researchers using the international juvenile dermatomyositis biopsy score tool.[2, 3] The scores assigned for muscle necrosis, regeneration, and inflammatory cell infiltration (CD4+T, CD8+T, and CD68+ macrophages) were categorized as 0 (none), 1 (mild), and 2 (high).

RNA extraction, library construction, and sequencing

Total RNA was extracted from muscle samples using Trizol reagent, and genomic DNA was subsequently removed using RNase-free TURBO DNase (Thermo Fisher Scientific). For library preparation, 1 μg of total RNA served as input material, utilizing the NEBNext Ultra RNA Library Prep Kit for Illumina (Catalog: #E7530L). Poly-A-containing mRNA from total RNA was isolated using poly-T oligo-attached magnetic beads. Fragmentation was performed using divalent cations under elevated temperature in the First Strand Synthesis Reaction Buffer (5X). First-strand cDNA synthesis was accomplished using random hexamer primers and M-MuLV Reverse Transcriptase, with concurrent degradation of RNA using RNase H. Subsequently, second-strand cDNA synthesis was conducted using DNA Polymerase I and dNTPs, and the remaining overhangs were transformed into blunt ends via exonuclease/polymerase activities. After adenylation of the 3' ends of the DNA fragments, adaptors with hairpin loop structures were ligated to facilitate hybridization. To selectively obtain cDNA fragments within the 370~420 bp range, the library fragments underwent purification using the AMPure XP system (Beckman Coulter). Then, PCR amplification of the library product was executed, followed by purification using AMPure XP beads. Finally, the libraries were sequenced on Illumina platforms using the PE150 strategy at Novogene Co., Ltd in Beijing, China.

Protein extraction and MS

For protein extraction, we employed a combination of cryogenic grinding and ultrasonication for muscle protein extraction, following these steps: (1) All muscle samples for RNA sequencing and MS analysis were stored in liquid nitrogen. (2) During protein extraction, muscle samples were cryogenically ground into powder using liquid nitrogen for 1 minute, repeated four times (Tissuelyser-24, Shanghai Jingxin Industrial Development Co., Ltd.). 1×SDS protein lysis buffer was immediately added and incubated for 30 minutes with a tube rotator at 4°C. (3) To aid tissue lysis, ultrasonication was performed using an 8-second sonication cycle followed by an 8-second resting period for 10 minutes at 4°C, utilizing a Scientz08-III Non-Touch Ultrasonic Homogenizer (Ningbo Scientz Biotechnology). After centrifugation at 12,000g for 20 minutes, the supernatant was collected for further analysis. The protein content was measured using the MicroBCA kit (Thermo Fisher Scientific).

For MS analysis, protein precipitation, dissolution, and enzymatic digestion into peptides were carried out using acetone, urea, and trypsin. LC-MS/MS analysis was performed by the DIA model with a NanoAcquity UHPLC coupled to an Orbitrap Exploris 240 mass spectrometer (Thermo Fisher Scientific). Briefly, 0.5μg peptides were enriched in a trap column at a flow rate of 10 μL/min for 4 min, and then separated on an analytical column (2 μm, Acclaim PepMap 100 C18 HPLC Columns, 100 Å , 250 x 0.075 mm) at a constant flow rate of 300 nL/min for 150 min with the following gradient: 0–3 min: 3% B; 3–5 min: 3–5% B; 5–125 min: 5–40% B, 125–135 min: 40–95% B, 135–139 min: 95% B; 139–140 min:

95%-3% B; 140–150 min: 3% B. Buffer A contained 0.1% Formic acid (FA) in HPLC water, while buffer B contained 80% acetonitrile (ACN) and 0.1% FA in HPLC water. The Captive Spray source was operated according to the manufacturer's recommendations: 2.1kV for the capillary voltage and 300 °C for the capillary temperature. A full MS scan was acquired (350-1200 m/z range) at a resolution of 60,000 using an AGC target value of 3e6 and maximum injection time of 20 ms. Following the full MS scan, MS2 scans with isolation windows of 10 m/z width spanning from 350-1200 m/z at 15,000 resolution were acquired. The normalized type of collision energy was selected and the collision energy of MS2 was set to 30%. Protein expression was quantified using DIA-NN (v1.8.1) with the human protein sequences from UniProt (www.uniprot.org/, Oct. 20th, 2022) as the library and precursor FDR < 0.05. DEP packages (v1.16.0) were selected to identify differentially expressed proteins (DEPs).

The quality control (QC) process for RNA sequencing and MS

Initially, a total of 211 IIM and 24 control muscles were included for QC. (1) For RNA-seq QC, the RNA abundance and quality were evaluated based on RNA amount, RNA concentration, RNA Integrity Number (RIN), A260/A280 ratio, and A260/230 ratio. Inclusion criteria were set at total RNA >1 μg, RNA concentration >120 μg/mL, RIN >8.0, 260/280 ratio between 1.8 and 2.0, and 260/230 ratio between 1.8 and 2.2. Subsequently, the quality of RNA sequences was assessed using fastQC (v0.12.0), considering metrics such as Q20 and Q30, GC content, length distribution, overrepresented sequences, and adapter content to filter out low-quality samples. (2) For MS QC, muscle samples with protein content > 2 μg/μL, total protein > 120 μg, and total peptide > 0.5 μg were used for MS analysis. Instrument condition evaluation involved injecting a QC sample (a mixed muscle lysis peptide sample) before each batch, and 22 muscle samples were analyzed a second time as biological replicates. In the data analysis phase, proteins with missing values exceeding 50% in each group were discarded. Finally, a total of 174 IIM (DM, IMNM, ASS) and 19 control muscles qualified for further analysis in both RNA sequencing and MS. Samples from the controls and each IIM subtype were age- and gender-matched (supplementary tables 1, 2). The flowchart of the QC process is presented in supplementary figure 1.

IR-related splicing factor analysis

To evaluate the association between splicing factors and IRs, Splicing factor-interacting Retained Intron Networks (SPIRONs) were constructed, following a previously reported method.[4] Briefly, the LASSO method with 5-fold cross-validation was employed to screen a subset of splicing genes that collectively explain the changes in IRs. Only IRs and splicing factors with $R^2 > 0.5$ and β score > 0.2 were selected. Then, a correlation network was built using Gephi (v0.10.1) to visualize the relationships among these selected IRs and splicing factors.

Real-time polymerase chain reaction (PCR)

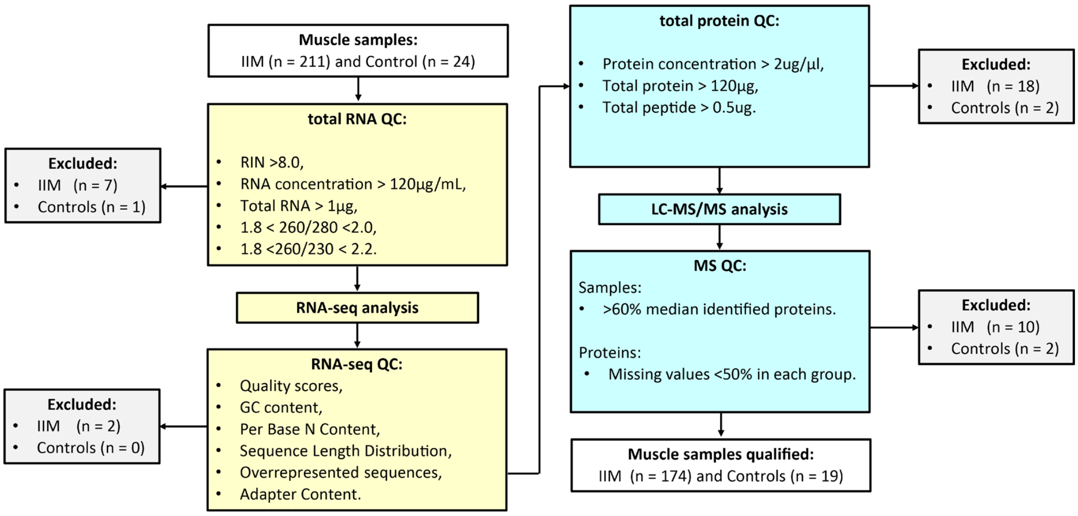
Reverse transcription and real-time PCR were carried out using The Evo M-MLV RT Kit and SYBR Green Pro Taq HS Premix (both from Accurate Biotechnology (Hunan) Co., Ltd, Changsha, China) on a qTOWER3 qPCR system (Analytik jena). The relative expression levels were calculated by the $2^{-\Delta\Delta CT}$ method and normalized to GAPDH. Details of the exon- and intron-specific primers are provided in supplementary table 4.

Western blot

For Western blotting, 40 μg protein aliquots were separated by 10% SDS-PAGE and transferred onto polyvinylidene difluoride membranes. The membranes were blocked with 5% nonfat dried milk in Tris-buffered saline with 0.1% Tween 20 for 2 h and then incubated with anti-human GAPDH, YBX1, and HSPA2 antibodies (all from Abcam, dilution: 1:1000) at 4 ℃ overnight. The following day, the membranes were washed and incubated with secondary antibodies. The blots were visualized using the enhanced chemiluminescence method (Bio-Rad) and the bands were quantified using ImageJ (v1.51).

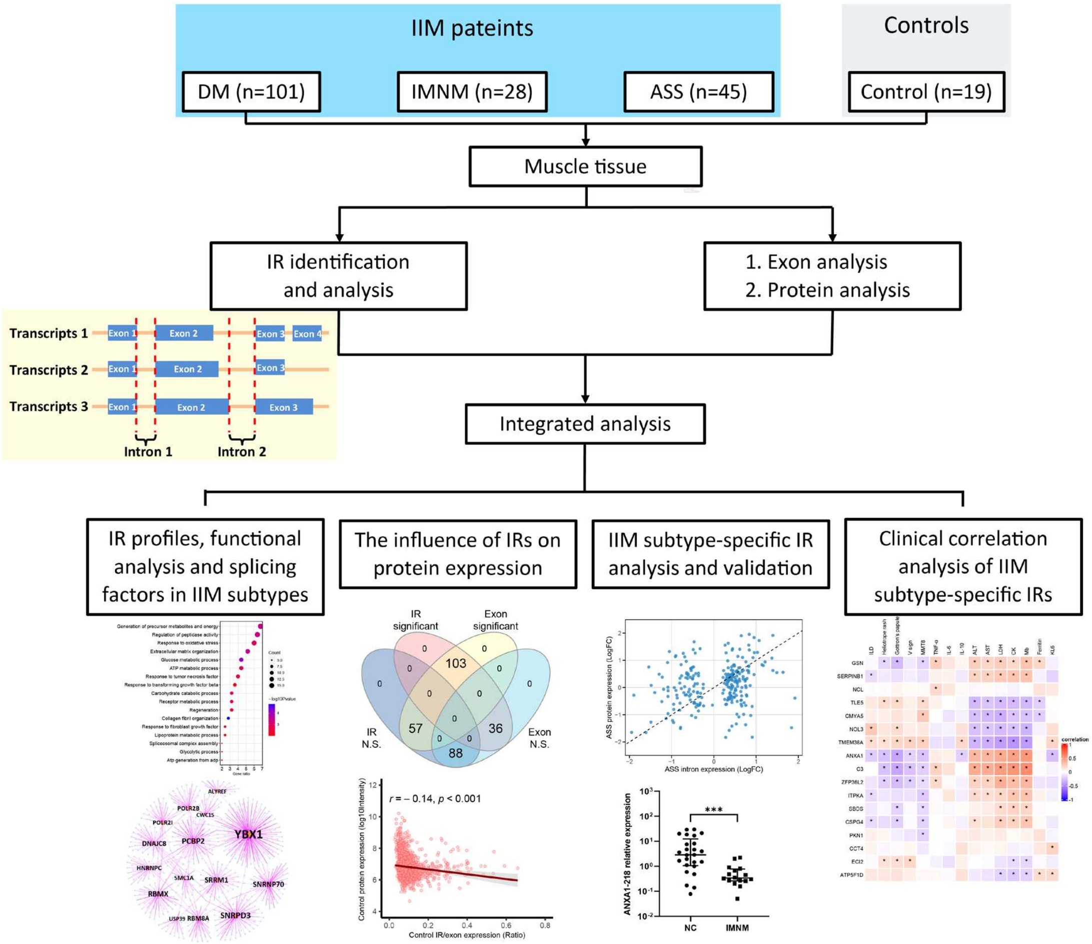
Supplementary Figures

Figure S1



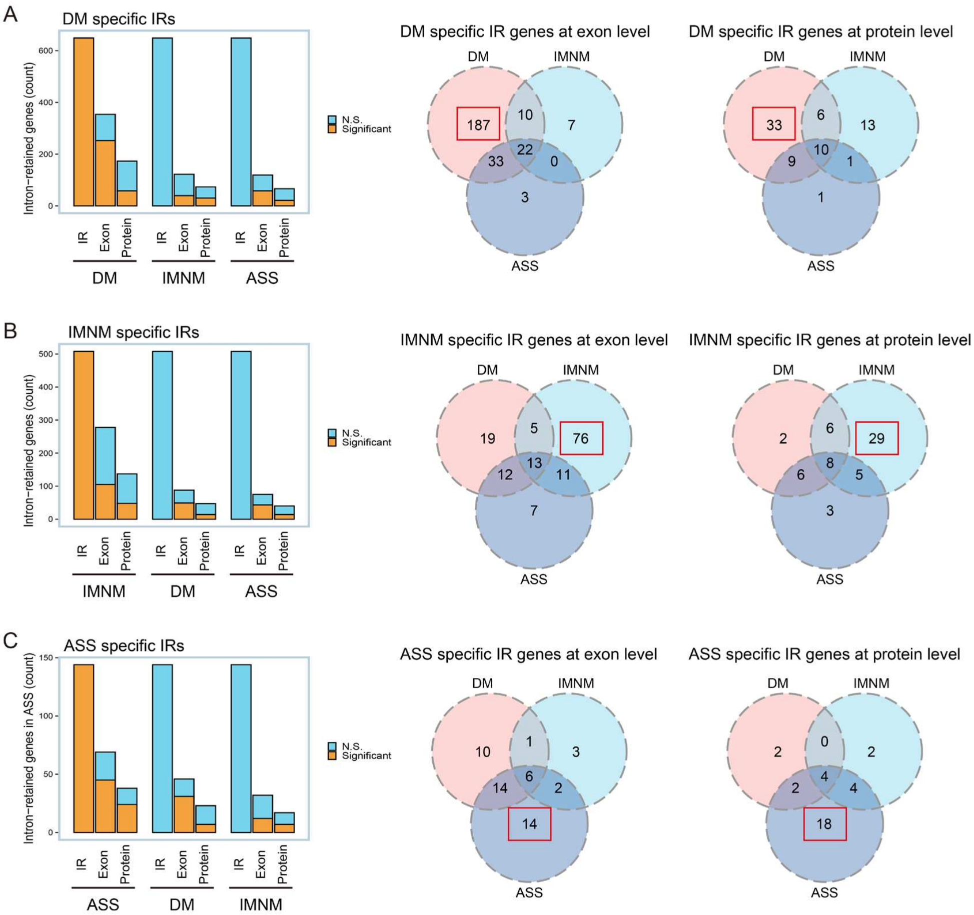
Supplementary Figure 1 Quality control flowchart of RNA sequencing and MS analysis. A total of 211 IIM and 24 control muscles were included for quality control (QC) procedures. For RNA-seq QC, criteria included total RNA >1 μg, RNA concentration >120 μg/mL, RIN >8.0, 260/280 ratio between 1.8 and 2.0, and 260/230 ratio between 1.8 and 2.2. Subsequently, RNA sequence quality was assessed using fastQC (v0.12.0), considering metrics such as Q20 and Q30, GC content, length distribution, overrepresented sequences, and adapter content to filter out low-quality samples. For MS QC, muscle samples with protein content > 2 μg/μL, total protein > 120 μg, and total peptide > 0.5 μg were selected for MS analysis. In the data analysis phase, proteins with missing values exceeding 50% in each group were discarded. During QC, 37 IIM and 5 control muscles were excluded. A total of 174 IIM and 19 control muscles qualified for further analysis in both RNA sequencing and MS.

Figure S2



Supplementary Figure 2 Overview of the outline and the work-flow of the study IR profiles were identified in muscle tissue from 174 IIM (101 DM, 28 IMNM, 45 ASS) and 19 controls. The study progressed through the following steps: (1) IIM-specific Differentially Expressed Introns (DEIs), their functions, and correlated splicing factors for IIM subtypes were analyzed by edgeR, Gene Ontology enrichment analysis, and Splicing Pathway-based Intron Regulation Network (SPIRON). (2) The influence of IRs on protein expression was investigated through an integration analysis of IR, exon, and protein data. (3) IIM subtype-specific IRs were identified and further validated using RT-PCR. (4) The correlation between IR expression and clinical manifestations, laboratory examination results, and muscle histopathology in IIM patients was assessed using point-biserial correlation or Spearman's correlation.

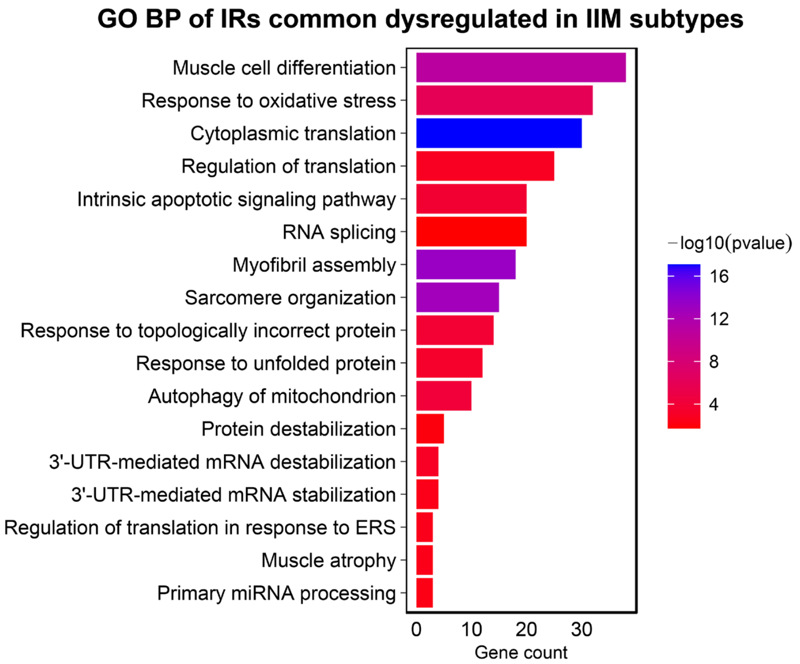
Figure S3



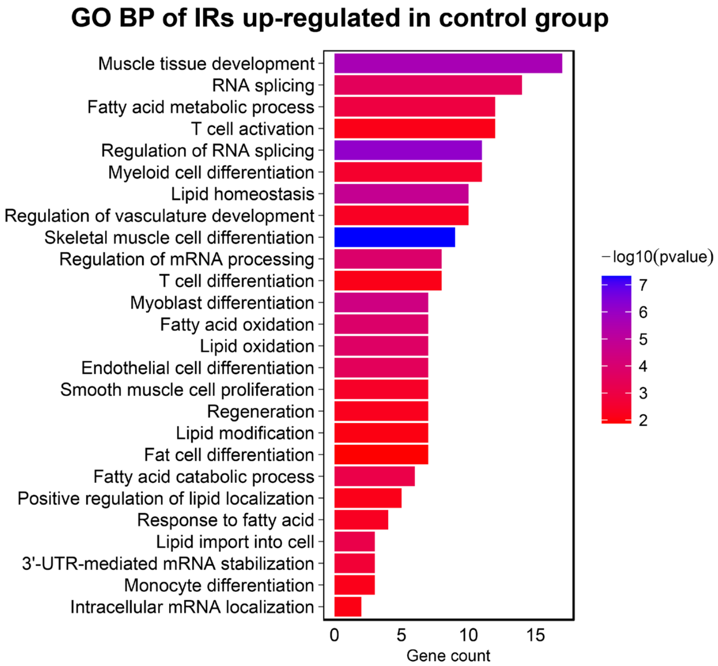
Supplementary Figure 3 The expression of exons and proteins for IIM subtype-specific IR genes across IIM subtypes. (**A**) The count of altered exons and proteins for DM subtype-specific IR genes in IIM subtypes (left panel). The Venn plot of DM-specific IR genes in IIM subtypes at the exon level (middle panel) or at the protein level (right panel). The red box indicates DM-specific IR genes specifically altered at the exon or protein levels in DM. (**B**) The count of altered exons and proteins for IMNM subtype-specific IR genes in IIM subtypes (left panel). The Venn plot of IMNM-specific IR genes in IIM subtypes at the exon level (middle panel) or at the protein level (right panel). The red box indicates IMNM-specific IR genes specifically altered at the exon or protein levels in IMNM. (**C**) The count of altered exons and proteins for ASS subtype-specific IR genes in IIM subtypes (left panel). The Venn plot of ASS-specific IR genes in IIM subtypes at the exon level (middle panel) or at the protein level (right panel). The red box indicates ASS-specific IR genes specifically altered at the exon or protein levels in ASS.

Figure S4

A

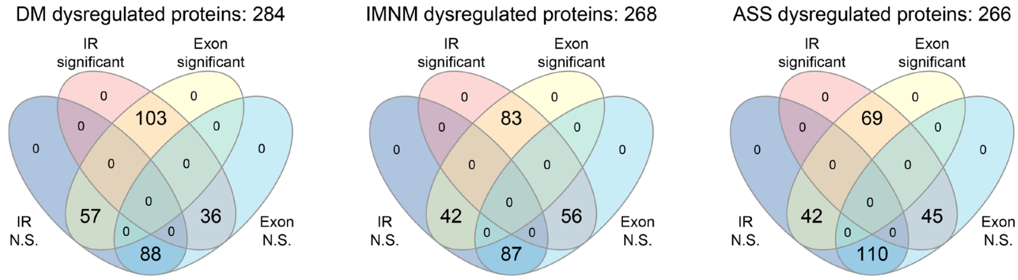


B



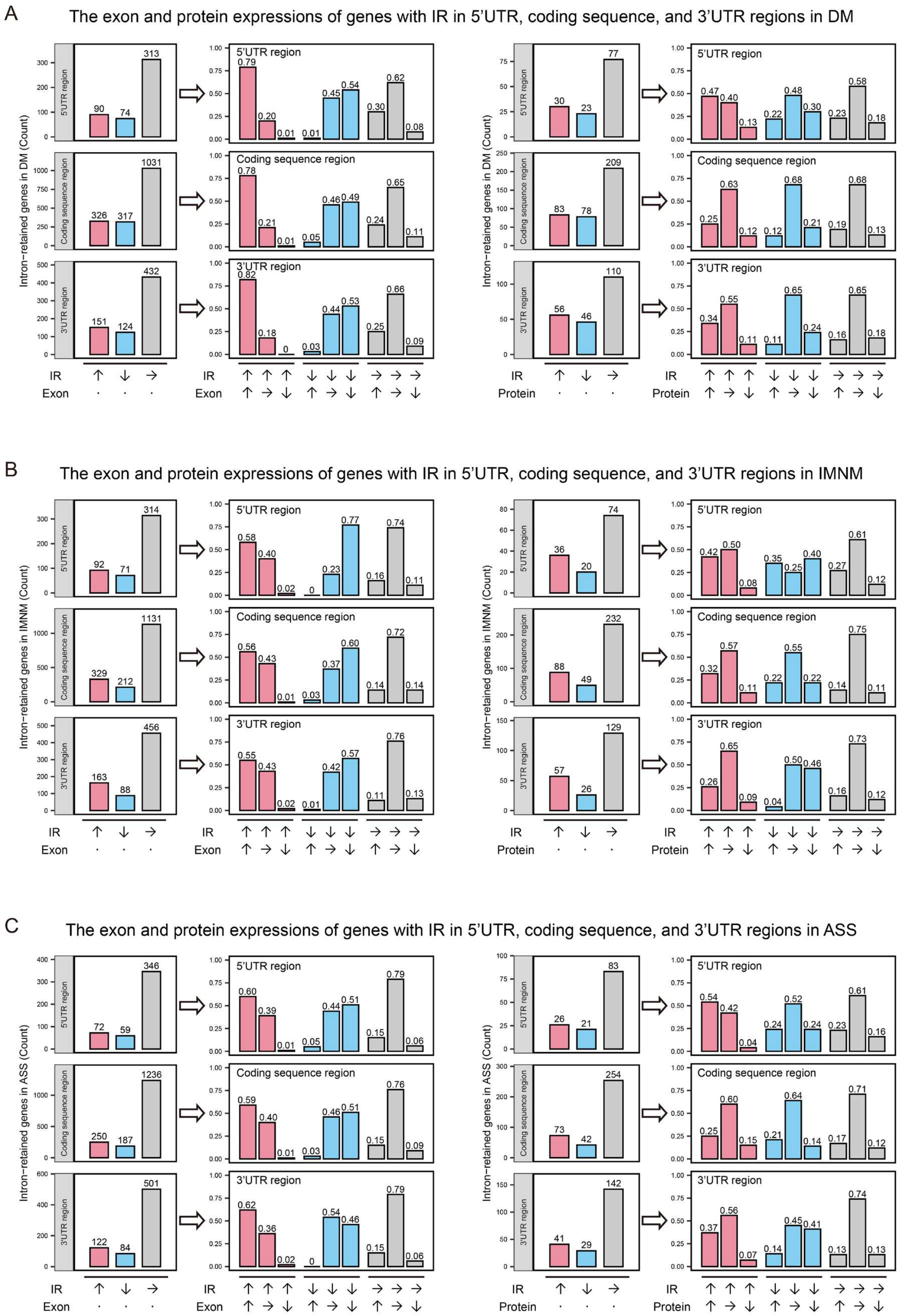
Supplementary Figure 4: Gene Ontology enrichment analysis of commonly altered IRs among IIM subtypes (A) or IR-related genes with increased IR levels in control muscles (B). The colors of the bars represent p-values, and the length of the bars represents the count of IR-related genes associated with the GO term.

Figure S5



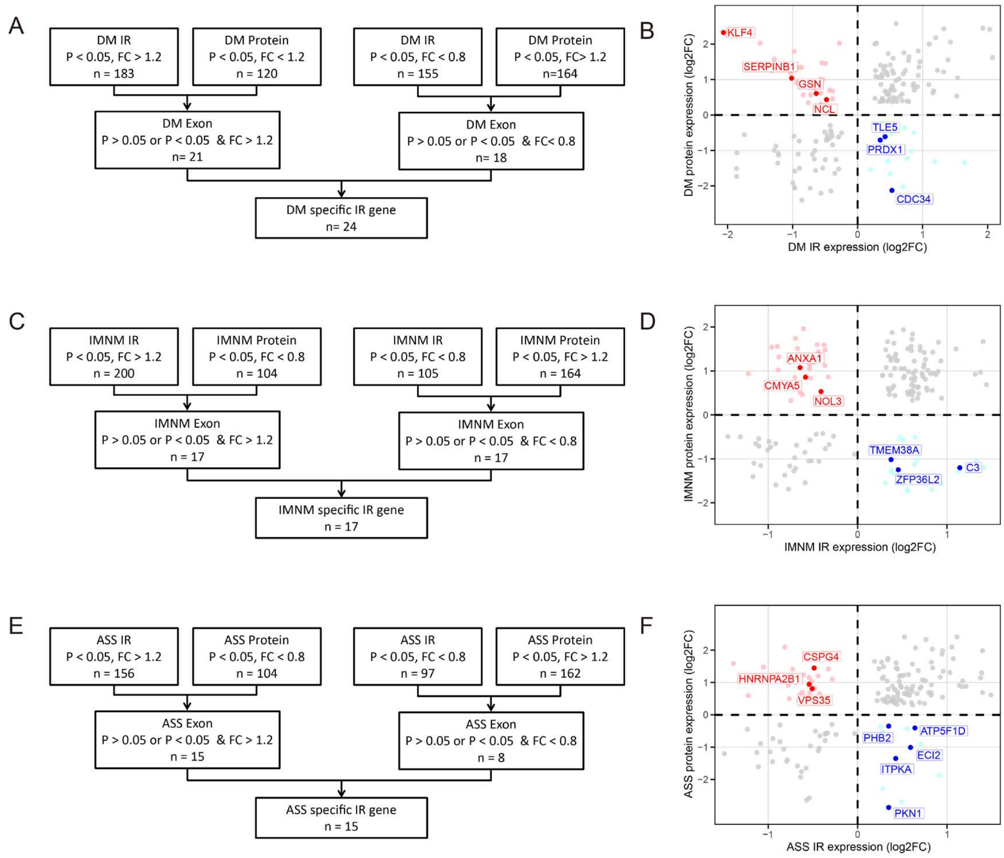
Supplementary Figure 5: IR and exon alterations in genes with dysregulated proteins of IIM subtypes. (**A**) The IR and exon alterations of dysregulated proteins in DM. (**B**) The IR and exon alterations of dysregulated proteins in IMNM. (**C**) The IR and exon alterations of dysregulated proteins in ASS. The red circle represents dysregulated proteins with IR alteration. The deep blue circle represents dysregulated proteins without IR alteration. The yellow circle represents dysregulated proteins with exon alteration. The light blue circle represents dysregulated proteins without exon alteration. (significant: FC > 1.2 or FC < 0.8, with P value < 0.05).

Figure S6


A
The exon and protein expressions of genes with IR in 5'UTR, coding sequence, and 3'UTR regions in DM
Intron−retained genes in DM (Count)
5'UTR region
Coding sequence region
3'UTR region
IR
Exon
Protein
B
The exon and protein expressions of genes with IR in 5'UTR, coding sequence, and 3'UTR regions in IMNM
Intron−retained genes in IMNM (Count)
C
The exon and protein expressions of genes with IR in 5'UTR, coding sequence, and 3'UTR regions in ASS
Intron−retained genes in ASS (Count)

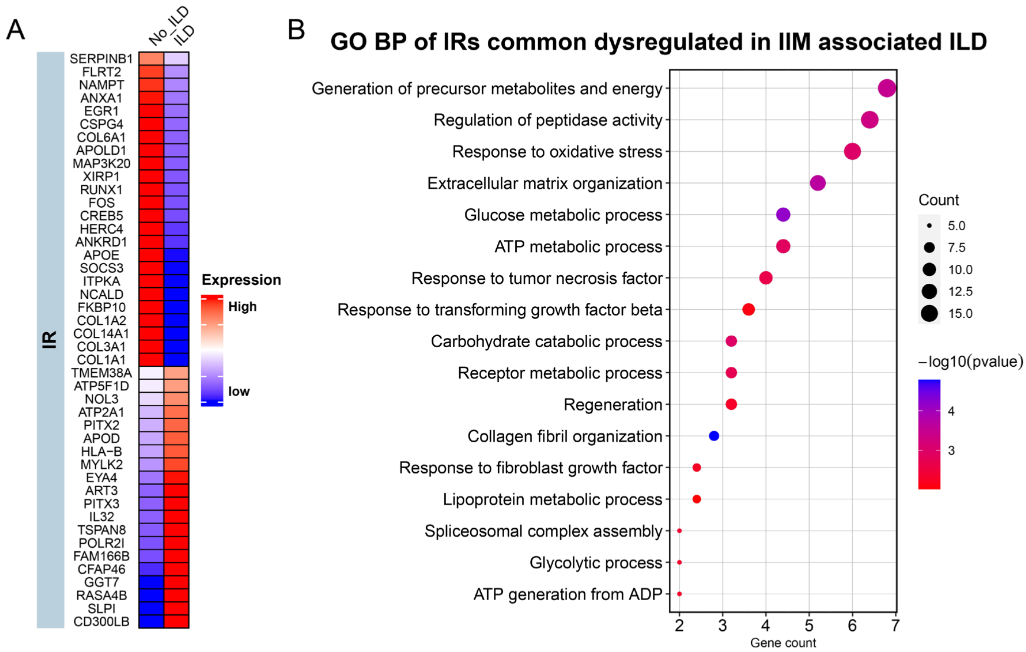
Supplementary Figure 6 The exon and protein expressions of genes with IRs in 5'UTR, coding sequence, and 3'UTR regions in DM (A), IMNM (B), and ASS (C). The IRs were categorized into 5'UTR, coding sequence, and 3'UTR groups based on their location in mRNA. Genes detected at both IR and exon or protein levels simultaneously were grouped according to their expression trends. The left panel displays the count of genes with upregulated, downregulated, or unchanged IRs. Subsequently, the right panel shows the percentage of exon or protein alteration among the genes with upregulated, downregulated, or unchanged IRs.

Figure S7



Supplementary Figure 7 Integration strategy of IR, protein, and exon analysis in DM, IMNM, and ASS. Genes were selected based on exhibiting either up-regulation at the IR level and down-regulation at the protein level, with no significant changes or up-regulation at the exon level; or decreased expression levels at the IR level and increased expression levels at the protein level, with no significant changes or an increase at the exon level. (**A**) In DM, 39 genes with opposing trends in IR and protein expression were identified, 24 of which were specific to DM. (**B**) The scatter plot illustrates genes with IR and protein alterations in DM, with specific IR genes in DM-related pathways highlighted. (**C**) In IMNM, 34 genes with opposing trends in IR and protein expression were identified, 17 of which were specific to IMNM. (**D**) The scatter plot illustrates genes with IR and protein alterations in IMNM, with specific IR genes in IMNM-related pathways highlighted. (**E**) In ASS, 23 genes with opposing trends in IR and protein expression were identified, 15 of which were specific to ASS. (**F**) The scatter plot illustrates genes with IR and protein alterations in ASS, with specific IR genes in ASS-related pathways highlighted. (FC: fold change)

Figure S8



Supplementary Figure 8 IR-related genes in IIM patients with interstitial lung disease (ILD) compared to IIM patients without ILD. (**A**). IR profiles in IIM-related ILD. (**B**) Functional enrichment analysis of the genes with altered IRs in IIM-related ILD. The colors of the dots represent p-values and the size of the dots represent the number of IR-related genes associated with the GO term.

Supplementary Tables

Supplementary Table 1. Comparison of clinical manifestations and laboratory data among IIM and controls for RNA sequencing and MS analysis

General information	DM	IMNM	ASS	Control	*P*
Total	101	28	45	19	
Age at onset(year)	49.44±10.75	45.90±10.41	49.23±11.39	47.63±15.49-	0.500
Male/female	32/69	8/20	14/31	7/12	0.949
Disease duration (month)	3.00 (2.00, 7.00)	4.00 (3.00, 8.00)	4.00 (2.00, 9.5)		0.237
ILD	61	11	36		0.002
Heliotrope rash	52	0	0		0.000
Gottron's sign	73	0	0		0.000
V sign	34	0	0		0.000
Mechanic's hand	10	1	4		0.571
Arthritis	40	4	23		0.007
MMT8	80.00 (68.00, 80.00)	67.00 (49.50, 80.00)	80.00 (71.50, 80.00)		0.002
ALT (U/L)	39.90 (23.20, 68.40)	172.40 (74.58, 263.15)	68.85 (31.15, 159.65)		0.000
AST (U/L)	48.90 (38.65, 97.08)	175.80 (87.25, 292.65)	70.40 (47.78, 145.28)		0.000
LDH (U/L)	352.00 (293.00, 466.50)	949.00 (687.00, 1295.50)	507.00 (358.80, 797.00)		0.000
CK (U/L)	131.30 (67.80, 464.00)	4569.75 (2715.65, 8909.10)	1259.50 (558.30, 3832.45)		0.000
C4 (mg/L)	252.00 (208.00. 354.50)	199.00 (168.00, 239.50)	184.50 (155.00, 217.50)		0.028
C3 (mg/L)	778.00 (630.00, 932.50)	800.50 (721.25, 906.50)	792.00 (684.50, 941.25)		0.669
IgG (g/L)	14.20 (11.45, 16.85)	14.05 (12.03, 16.58)	16.15 (12.28, 22.25)		0.025
IgA (g/L)	2620.00 (1880.00, 3390.00)	2860.00 (1927.50, 3652.50)	2475.00 (1942.50, 3692.50)		0.815
IgM (g/L)	1320.00 (920.00, 1970.00)	1470.00 (972.00, 2047.50)	1680.00 (1110.00, 2860.00)		0.101
CRP (mg/L)	3.73 (2.12, 8.04)	2.92 (2.03, 6.40)	13.10 (5.76, 28.10)		0.000
ESR (mm/h)	42.00 (26.50, 64.00)	48.50 (29.50, 56.00)	58.00 (33.50, 87.00)		0.001
IL-1β (pg/ml)	5.00 (5.00, 7.82)	5.32 (5.00, 13.45)	5.00 (5.00, 9.36)		0.259
IL-6 (pg/ml)	5.06 (2.29, 11.35)	5.31 (2.65, 11.65)	9.31 (4.33, 15.90)		0.496
TNF-α (pg/ml)	12.30 (6.80, 16.60)	10.95 (6.82, 14.20)	18.60 (12.70, 20.85)		0.667
Anti-Ro52	51	16	33		0.287
Anti-SSA	14	8	15		0.017
MSAs	MDA5+: 57	SRP+: 20	Jo-1+: 25		
	TIF1γ+: 20	HMGCR+: 8	PL-7+: 7		
	SAE+: 6		EJ+: 9		
	NXP-2+: 8		PL-12+: 4		
	Mi2+: 10				

DM dermatomyositis; IMNM immune-mediated necrotizing myopathy, ASS antisynthetase syndrome, ILD interstitial lung disease, ALT alanine transaminase, AST aspartate aminotransferase, LDH lactate dehydrogenase, CK creatine kinase, Mb Myoglobin, C4 complement 4, C3 complement 3, IgG immunoglobulin G, IgA immunoglobulin A, IgM immunoglobulin M, CRP C-reactive protein, ESR erythrocyte sedimentation rate, MSA myositis-specific antibody

Supplementary Table 2. Comparison of clinical manifestations and laboratory data among IIM and controls in RT-PCR validation

General information	**DM**	**IMNM**	**ASS**	**Control**	***P***
Total	53	16	15	27	
Age at onset(year)	51.5±15.68	49.50±14.89	49.00±14.99	50.49±17.63	0.931
Male/female	17/35	5/11	4/12	9/18	0.844
Disease duration (month)	3.00 (2.00, 6.00)	3.00 (2.25, 5.75)	3.00 (1.50, 6.50)		0.855
ILD	32	11	6		0.151
Heliotrope rash	26	0	0		0.000
Gottron’s sign	36	0	0		0.000
V sign	18	0	0		0.000
Mechanic’s hand	8	2	1		0.669
Arthritis	24	5	3		0.146
MMT8	80.00 (68.00, 80.00)	67.00 (49.50, 80.00)	80.00 (71.50, 80.00)		0.937
ALT (U/L)	40.90 (23.40, 106.80)	108.40 (15.50, 257.00)	68.20 (16.90 ,157.80)	-	0.142
AST (U/L)	52.40 (40.20, 106.80)	173.30 (34.70, 274.60)	85.40 (48.70 ,131.80)	-	0.070
LDH (U/L)	352.00 (299.00, 444.00)	786.00 (680.00, 1274.30)	455.00 (378.00, 785.00)	-	0.000
CK (U/L)	157.20 (88.40, 580.00)	3021.40 (93.10, 5213.00)	1090.40 (645.20, 2949.10)	-	0.000
C4 (mg/L)	255.00 (213.00, 678.00)	177.50 (141.25, 247.75)	190.00 (161.00, 299.00)		0.282
C3 (mg/L)	717.00 (322.00,926.00)	689.00 (582.30, 811.00)	735.00 (455.00,895.00)		0.946
IgG (g/L)	15.00 (11.50, 17.60)	15.10 (11.83, 22.55)	16.20 (12.90 ,26.50)	-	0.168
IgA (g/L)	2610.00 (1830.00, 3530.00)	2445.00 (1647.50, 3430.00)	2260.00 (1680.00, 3990.00)		0.454
IgM (g/L)	1320.00 (949.00, 2010.00)	1760.00 (1270.00, 2120.00)	2900.00 (1270.00, 3220.00)		0.011
CRP (mg/L)	3.47 (2.07, 8.22)	2.82 (2.15, 11.10)	10.60 (6.40, 41.80)	-	0.008
ESR (mm/h)	48.00 (28.00, 69.00)	44.00 (22.00, 88.00)	73.00 (46.00, 112.00)	-	0.042
IL-1β (pg/ml)	5.00 (5.00, 7.13)	9.45 (5.21, 26.53)	5.00 (5.00, 7.87)	-	0.026
IL-6 (pg/ml)	6.60 (2.63, 12.50)	5.41 (2.07, 11.32)	10.40 (4.21, 23.58)		0.462
TNF-α (pg/ml)	13.00 (10.20, 18.20)	16.00 (6.97, 24.70)	19.50 (16.30, 21.75)		0.974
Anti-Ro52	26	11	14		0.184
Anti-SSA	7	2	4		0.463
MSAs	MDA5+: 35	SRP+: 11	Jo-1+: 8		
	TIF1γ+: 8	HMGCR+: 5	PL-7+: 4		
	SAE+: 1		EJ+: 2		
	NXP-2+: 5		PL-12+: 1		
	Mi2+: 4				

DM dermatomyositis; IMNM immune-mediated necrotizing myopathy, ASS antisynthetase syndrome, ILD interstitial lung disease, ALT alanine transaminase, AST aspartate aminotransferase, LDH lactate dehydrogenase, CK creatine kinase, Mb myoglobin, C4 complement 4, C3 complement 3, IgG immunoglobulin G, IgA immunoglobulin A, IgM immunoglobulin M, CRP C-reactive protein, ESR erythrocyte sedimentation rate, MSA myositis-specific antibody

Supplementary Table 3. Comparison of histopathological features among IIM subtypes for RNA sequencing and MS analysis

	DM	IMNM	ASS	*P*
Total	101	28	45	
Muscle fiber necrosis	36	28	32	0.000
Mild necrosis	20	6	14	0.317
Severe necrosis	16	22	18	0.000
Perifascicular necrosis	7	0	12	0.000
Perifascicular atrophy	20	0	4	0.015
Regenerating fiber	31	25	28	0.000
Vascultis	18	5	13	0.288
Connective tissue proliferation	72	23	31	0.420
COX negative fiber	6	0	3	0.394
MHC-I expression on sarcolemma	72	18	32	0.763
Focal expression	53	13	23	0.852
Diffuse expression	19	5	9	0.055
CD4+ T cell	47	17	19	0.287
Endomysia	25	14	13	0.035
Perimysium	34	5	14	0.273
CD8+ T cell	35	12	24	0.135
Endomysia	17	11	16	0.008
Perimysium	23	3	13	0.192
CD68+ macrophage	59	24	30	0.146
Endomysia	38	20	26	0.002
Perimysium	38	13	17	0.088
CD20+ B cell	14	1	4	0.160
Endomysia	5	1	2	0.937
Perimysium	14	1	3	0.177
MAC	28	17	11	
Sarcolemma of non-necrotic muscle fiber	7	14	9	0.000
Capillaries	28	3	3	0.006

DM dermatomyositis; IMNM immune-mediated necrotizing myopathy, ASS antisynthetase syndrome, COX cytochrome c oxidase, MHC major histocompatibility complex, MAC membrane attack complex

Supplementary Table 4. Sequences of specific primers used in this study

Gene	Groups	Forward primer (5'–3')	Reverse primer (5'–3')
GADPH	Exon	GTCTCCTCTGACTTCAACAGCG	ACCACCCTGTTGCTGTAGCCAA
CDC34-86	IR	CGCACAGGGCTTTGTCCTAA	GGGGACAGGGTGGGTGAG
SERPINB1-1631	IR	AACCATGCCCGAAGCCAA	TGTTAGTGAGTAGGGCTTTCC
SERPINB1-1732	IR	AGACGCTCCACGTTTGTTGA	AGAAGGAAACACCAAGGAATCACT
ANXA1-3311	IR	TGGGAGATGAGAGGTGAAGAA	GCTGGTTTGCTTGTGGCG
ANXA1-1304	IR	ACAAAGTTCTGGACCTGGAGT	AACTCTTTCAACTGCTGGACT
ANXA1-228	IR	CACGGGAAGGGTGATGTACT	CAACAAATCAGACTCATAAAGTGGG
C3-41	IR	CAGGGCTTCTCTGCACTTGA	CTCCCGTGGCTCTTCCCT
C3-173	IR	TTGTCCAGCCGTTCTTCCAG	GGCAGTAGGCCCCAGATAAG
CMYA5-2908	IR	GCATGGAGGAGCCACAAGAT	TGGTTGTTGGGGTTTTGTTGT
ATP5F1D-702	IR	TGGACTTGACCAGGGGAGTC	GCCAACAACTGCACCGAAG
ECI2-1332	IR	CCCAACCCTCACATTTCTGC	AGCCAGTTTGCTGAGGCAAG
HNRNPA2B1-2017	IR	ATAACCGGGGCTACCTCCAA	AGCTGGAAATGGATGTGAGGA
HNRNPA2B1-160	IR	TGCTGATAGTTATTTCCTCCATGA	ACTACGAACAATGGGGAAAGC
KLF4-1101	IR	GCGAATTTCCATCCACAGCC	AGGATAGGGTCAGAGTTGGT
KLF4-482	IR	CCCGACATACTGACGTGCTG	TGGGCAGACAGGACATAACAC
SERPINB1	Exon	AGCTCAGCATGGTCATCCTGCT	CGAGATTCTCAGGTTTAGTCCAC
ATP5F1D	Exon	CGGAGCCTTCGGCATCCTGG	AAGAGTCGGCGTTCACTGCGAT
C3	Exon	GTGGAAATCCGAGCCGTTCTCT	GATGGTTACGGTCTGCTGGTGA
HNRNPA2B1	Exon	CAGCAACCTTCTAACTACGGTCC	CACTGCCTCCTGGACCATAGTT
KLF4	Exon	CATCTCAAGGCACACCTGCGAA	TCGGTCGCATTTTTGGCACTGG
CMYA5	Exon	GAACTTCACTGGATGTAGCCTGC	CCATCGGATAGTGGCTGTGTTC
ANXA1	Exon	GCGAAACAATGCACAGCGTCAAC	CAACCTCCTCAAGGTGACCTGT
ECI2	Exon	GCTGCCAGCAAGGATGACTCAA	GCTTTCTCCTCTACTCCACCAG
CDC34	Exon	TACTACGAGGGCGGCTACTTCA	GTGCCACATCTTGGTCAGGAAC

Supplementary Table 5 The IR-related genes in DM muscles

gene_symbol	Exon logFC	Exon p value	IR logFC	IR p value	Protein logFC	Protein p value
APEH	0.334	0.002	0.378	0.002	-0.280	0.014
ATP5PF	0.264	0.032	0.615	0.002	-0.469	0.032
CDC34	0.228	0.036	0.528	0.000	-1.814	0.000
CDH5	-0.506	0.000	-1.314	0.000	1.192	0.029
COX17	0.592	0.000	0.608	0.006	-0.664	0.037
COX5A	0.342	0.018	0.521	0.007	-0.566	0.004
DYNLL1	0.015	0.912	0.700	0.002	-1.899	0.000
FABP3	-0.423	0.053	-0.837	0.000	0.518	0.007
GSN	-0.638	0.000	-0.636	0.000	0.560	0.000
HMBOX1	-0.366	0.000	-0.495	0.001	1.463	0.000
IGHG1	3.079	0.002	0.865	0.047	-0.547	0.023
KLF4	-2.165	0.000	-2.063	0.000	2.534	0.000
KRT10	1.357	0.000	0.364	0.044	-0.900	0.001
MYBPC1	0.029	0.858	-0.386	0.009	0.626	0.000
NAMPT	-0.364	0.024	-1.028	0.000	1.059	0.000
NCL	-0.002	0.974	-0.478	0.006	0.384	0.007
NDUFB3	0.092	0.561	0.335	0.025	-0.912	0.008
PRDX1	0.201	0.037	0.348	0.021	-0.756	0.000
PSMD3	0.565	0.000	0.523	0.000	-0.554	0.047
SERPINB1	-0.626	0.000	-1.014	0.000	0.989	0.000
SLIRP	0.522	0.000	0.498	0.016	-1.429	0.002
TFIP11	0.557	0.000	0.272	0.041	-0.707	0.045
TLE5	0.678	0.000	0.422	0.000	-0.638	0.039
TRMT13	-0.084	0.297	-0.281	0.015	2.055	0.000

Supplementary Table 6 The IR-related genes in IMNM muscles

gene_symbol	Exon logFC	Exon p value	IR logFC	IR p value	Protein logFC	Protein p value
ANXA1	0.027	0.895	-0.643	0.033	1.003	0.011
APOE	1.173	0.005	1.327	0.002	-1.174	0.025
ATP5F1C	0.002	0.985	0.482	0.009	-0.343	0.027
C3	0.867	0.001	1.143	0.000	-1.274	0.000
CMYA5	0.255	0.205	-0.583	0.023	0.785	0.005
GANAB	0.067	0.317	-0.328	0.018	0.792	0.013
KIF1C	0.238	0.122	0.309	0.033	-1.431	0.036
MBNL1	-0.413	0.001	-0.898	0.000	1.509	0.005
MCTS1	0.517	0.139	0.667	0.030	-1.027	0.044
NOL3	-0.300	0.028	-0.409	0.010	0.459	0.048
NPTN	-0.355	0.000	-0.695	0.000	1.511	0.019
PM20D2	-0.003	0.975	-0.375	0.044	1.401	0.016
RRAD	0.232	0.365	-0.633	0.033	1.113	0.046
SFPQ	-0.038	0.664	-0.538	0.000	1.010	0.001
TMEM38A	0.234	0.168	0.374	0.042	-1.130	0.014
XIRP1	-1.311	0.000	-3.414	0.000	0.729	0.001
ZFP36L2	-0.180	0.225	0.455	0.020	-1.113	0.043

Supplementary Table 7 The IR-related genes in ASS muscles

gene_symbol	Exon logFC	Exon p value	IR logFC	IR p value	Protein logFC	Protein p value
ATP5F1D	0.280	0.104	0.641	0.000	-0.473	0.009
CSPG4	-0.043	0.854	-0.487	0.034	1.397	0.017
DDX3X	-0.190	0.038	-0.556	0.002	0.380	0.034
ECI2	0.499	0.002	0.591	0.048	-1.077	0.006
EIF4H	0.190	0.058	-0.480	0.042	0.761	0.000
HNRNPA2B1	0.098	0.240	-0.543	0.000	0.704	0.009
IGHA1	4.204	0.000	1.840	0.000	-1.477	0.000
ITPKA	1.367	0.080	0.426	0.041	-1.418	0.006
MAP7D2	1.346	0.035	0.545	0.007	-0.977	0.039
MPC1	0.246	0.051	0.281	0.043	-0.621	0.037
PHB2	0.252	0.028	0.348	0.001	-0.416	0.011
PKN1	0.088	0.401	0.348	0.021	-2.726	0.000
SYNPO2	-0.015	0.925	-0.609	0.002	0.449	0.002
TARBP1	-0.145	0.122	-0.441	0.001	1.265	0.038
VPS35	-0.099	0.157	-0.508	0.008	0.742	0.013

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